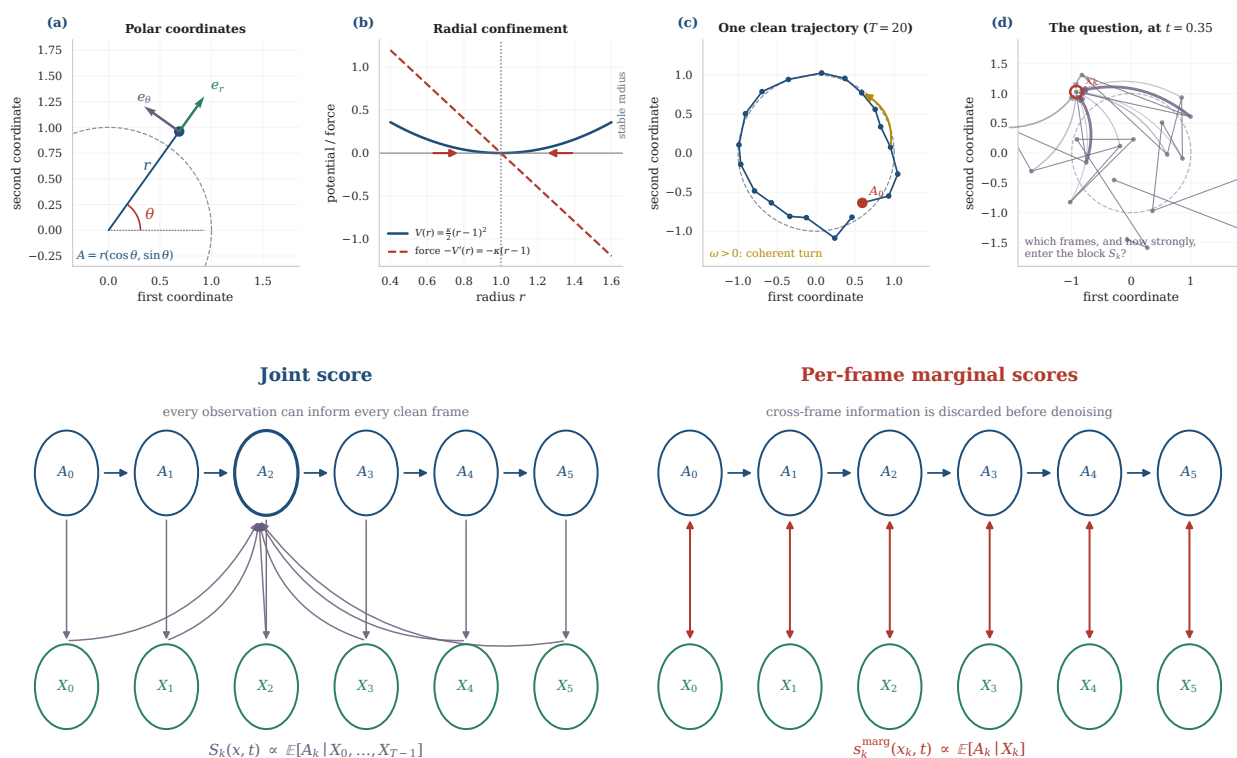


Joint Score Dynamics on a Rotating Ring

The first board problem: from an exactly solvable surrogate to the polar process



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26 July 2026

This note develops the first of the three problems discussed on the board with Marc Mézard. It reconstructs the background it needs from first principles, derives every identity it uses, separates the exactly solvable surrogate from the polar process that was actually drawn, and reports what the simulations do and do not establish.

Abstract

A dynamic object is a whole trajectory, not a collection of independent frames. This note studies the joint diffusion score of a noisy rotating trajectory near a unit ring, and asks how the Markov structure of the clean trajectory can be used to compute and interpret that score. The work proceeds in two stages. A Cartesian rotating-random-walk surrogate is solved in closed form: the known rotation is removed by an orthogonal change of coordinates, after which the noised score is a Gaussian random-walk term plus a rank-one correction generated by the shared ring anchor. The polar process of the board is then treated directly: its radial and angular one-step transitions are derived from the stochastic differential equations, its noised score is written as a posterior smoothing mean and evaluated by forward–backward integration on a polar grid, and its response to perturbing another frame is shown to equal a posterior covariance. A first-order expansion around one rotating branch gives an interpretable linear-Gaussian approximation, together with an explicit statement of when it may be used. Three diagnostics are then measured — the absolute intensity of cross-frame influence, its temporal reach, and the finite window needed to reproduce the score — and a fourth is introduced that combines reach and intensity in a single bounded number. In the one parameter set studied here, increasing diffusion time broadens the normalised reach while strongly reducing the absolute intensity, and tangential information travels farther than radial information at low noise. These are measurements in a small, specific model, not general laws.

What the note concludes

1. The clean trajectory is Markov and its clean score is local, but the score of the *noised* joint trajectory is not local, in either model studied.
2. Clean Markovity remains useful in two concrete ways: the posterior is computable by repeated one-step operations rather than one integral over $2T$ latent variables, and any distant influence must be transmitted through a chain of local transitions, which gives the nonlocality a structured profile.
3. The cross-frame response of the score is exactly a posterior covariance, $\partial S_k / \partial x_j = -\delta_{kj} \Delta_t^{-1} I_2 + m_t^2 \Delta_t^{-2} \text{Cov}(A_k, A_j | x)$. This is verified numerically to 6.6×10^{-9} .
4. Reach, intensity, and functional receptive field are different observables and move in different directions with diffusion time. Reporting only a normalised correlation length is misleading in a finite chain.
5. At the parameters studied, tangential information reaches much farther than radial information at low noise. At high noise both channels are broad but weak and the score approaches $-x$.
6. The local linear model reproduces the score to a few per cent while the posterior stays within one arc of the circle; it cannot represent angular wrapping or a multimodal phase.

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1 Background: the tools this note uses

The problem below is a small one, but reading it requires stochastic calculus and a few ideas from statistical mechanics. This section assembles exactly those, and no more. Readers who already have them can go to [Section 2](#).

1.1 Why diffusion models

Diffusion models are the dominant approach to generative modelling of continuous data. They underlie the image generators deployed by the large technology companies — latent diffusion in Stable Diffusion ([Rombach et al., 2022](#)), the DALL·E and Imagen families, Adobe Firefly — and the same construction, with a transformer denoiser, drives video generators such as Sora and Veo. The reach extends past media: AlphaFold 3 predicts atomic coordinates with a diffusion module ([Abramson et al., 2024](#)), and protein design pipelines such as RFdiffusion generate backbones the same way. What all of these share is a single mechanism: corrupt data by a known noising process, learn to reverse it, and sample by iterated denoising.

The mechanism is attractive for a theorist because the object being learned is not an arbitrary function. It is the *score* $\nabla_x \log p_t(x)$ of a family of smoothed data distributions, and that family is generated by a linear stochastic differential equation. This makes diffusion models unusually open to exact analysis, which is what the present note exploits: in a small model the score can be computed to machine precision and its structure inspected directly, rather than estimated from a trained network.

1.2 Brownian motion, and why ordinary calculus fails

Brown observed in 1827 that pollen grains suspended in water jitter without pause. [Einstein \(1905\)](#) explained it quantitatively: the jitter is the accumulated effect of molecular collisions, the mean square displacement grows linearly in time, $\mathbb{E}[\|X_u - X_0\|^2] = 2Du$, and the diffusivity is tied to temperature and friction by $D = k_B T / \gamma$. Wiener later constructed the limiting process rigorously; we call it Brownian motion, or the Wiener process B_u , defined by $B_0 = 0$, independent increments, $B_v - B_u \sim \mathcal{N}(0, v - u)$, and continuous paths.

The linear growth of the variance is the source of every technical difficulty that follows. An increment scales as $dB \sim \sqrt{du}$, which is much larger than du for small du . Brownian paths are therefore nowhere differentiable, and $\int f dB$ cannot be defined the way a Riemann–Stieltjes integral is defined.

Toolbox 1. Total variation and quadratic variation

Fix $[0, u]$ and a partition $0 = s_0 < \dots < s_n = u$ of mesh $\rightarrow 0$. Two sums behave very differently.

Quadratic variation is finite. Let $Q_n = \sum_i (B_{s_{i+1}} - B_{s_i})^2$. Each increment is $\mathcal{N}(0, \Delta s_i)$ with $\Delta s_i = s_{i+1} - s_i$, so

$$\mathbb{E}[Q_n] = \sum_i \Delta s_i = u, \quad \text{Var}(Q_n) = \sum_i \text{Var}[(B_{s_{i+1}} - B_{s_i})^2] = 2 \sum_i (\Delta s_i)^2,$$

using $\text{Var}(Z^2) = 2\sigma^4$ for $Z \sim \mathcal{N}(0, \sigma^2)$. Since $\sum_i (\Delta s_i)^2 \leq \max_i \Delta s_i \cdot u \rightarrow 0$, we get $Q_n \rightarrow u$ in mean square. In shorthand,

$$(dB_u)^2 = du.$$

Total variation is infinite. Let $V_n = \sum_i |B_{s_{i+1}} - B_{s_i}|$. Then

$$Q_n \leq \left(\max_i |B_{s_{i+1}} - B_{s_i}| \right) \cdot V_n.$$

Brownian paths are continuous, so the maximum increment tends to 0 as the mesh does. If V_n stayed bounded, the right-hand side would tend to 0, contradicting $Q_n \rightarrow u > 0$. Hence $V_n \rightarrow \infty$: Brownian paths have unbounded total variation on every interval.

Consequence. The Riemann–Stieltjes integral $\int f dg$ requires g to have bounded variation, precisely so that the limit does not depend on where inside each subinterval f is evaluated. For Brownian motion that guarantee is gone, and different evaluation points give genuinely different limits. A convention must be chosen. $\square$

Toolbox 2. The Itô integral, its mean, and its variance

Itô's convention is to evaluate the integrand at the *left* endpoint of each subinterval (Itô, 1944):

$$\int_0^u f_s dB_s := \lim_{n \rightarrow \infty} \sum_i f_{s_i} (B_{s_{i+1}} - B_{s_i}), \quad (1)$$

the limit taken in mean square, first for piecewise-constant adapted f and then extended by density to all adapted f with $\mathbb{E} \int_0^u f_s^2 ds < \infty$.

The choice of the left endpoint is what makes the construction useful. Because f_{s_i} is determined by information available at time s_i , and the increment $B_{s_{i+1}} - B_{s_i}$ is independent of that information with mean zero,

$$\mathbb{E}[f_{s_i} (B_{s_{i+1}} - B_{s_i})] = \mathbb{E}[f_{s_i}] \mathbb{E}[B_{s_{i+1}} - B_{s_i}] = 0, \quad (2)$$

so the integral has mean zero; it is a martingale. For $i < j$ the same argument applied to the later increment gives $\mathbb{E}[f_{s_i} \Delta B_i f_{s_j} \Delta B_j] = 0$, so the cross terms in the second moment vanish and

$$\mathbb{E} \left[\left(\int_0^u f_s dB_s \right)^2 \right] = \sum_i \mathbb{E}[f_{s_i}^2] \Delta s_i \longrightarrow \int_0^u \mathbb{E}[f_s^2] ds. \quad (3)$$

This is the *Itô isometry*. Finally, if f is *deterministic*, each term of (1) is a fixed multiple of an independent Gaussian, so the integral is Gaussian with mean 0 and variance $\int_0^u f_s^2 ds$. Every stochastic integral in this note is of that deterministic type. $\square$

The price of the convention is a modified chain rule. This is the single formal difference between an ordinary differential equation and a stochastic one: an ODE $\dot{X}_u = b(X_u)$ transports a point along a velocity field, and any smooth function of the solution obeys $d\phi = \phi' dX$; an SDE $dX_u = b(X_u) du + \sigma(X_u) dB_u$ carries a term of order $\sqrt{du}$, whose square is of order du and therefore cannot be dropped.

Toolbox 3. Itô's lemma and the product rule

Let X_u solve $dX_u = b du + \sigma dB_u$ and let $\phi(x, u)$ be twice differentiable. Expand to second order and substitute:

$$d\phi = \partial_u \phi du + \partial_x \phi dX + \frac{1}{2} \partial_x^2 \phi (dX)^2 + \dots, \quad (dX)^2 = b^2 (du)^2 + 2b\sigma du dB + \sigma^2 (dB)^2.$$

The Itô multiplication table — $(du)^2 = 0$, $du dB = 0$, $(dB)^2 = du$, the last being [Section 1.2](#) — leaves

$$d\phi = \left(\partial_u \phi + b \partial_x \phi + \frac{\sigma^2}{2} \partial_x^2 \phi \right) du + \sigma \partial_x \phi dB_u. \quad (4)$$

The half-second-derivative term is absent from ordinary calculus. It is why densities spread, why the Fokker–Planck equation is second order, and why exponential substitutions in SDEs must be checked rather than assumed.

The case used repeatedly below. If $\phi(x, u) = g(u)x$ is *linear* in x then $\partial_x^2 \phi = 0$, the correction vanishes, and (4) reduces to the familiar product rule

$$d(g(u)X_u) = g'(u)X_u du + g(u) dX_u. \quad (5)$$

Every integrating factor in this note is of this form, so no Itô correction ever appears — but that is a fact to be verified, not assumed. $\square$

1.3 Langevin dynamics, Fokker–Planck, and the Boltzmann law

[Langevin \(1908\)](#) wrote Newton’s equation for a particle in a fluid with the collisions represented as noise,

$$m\ddot{X}_u = -\gamma\dot{X}_u - V'(X_u) + \sqrt{2\gamma k_B T} \dot{B}_u. \quad (6)$$

When friction dominates inertia, the velocity relaxes fast and can be eliminated; the resulting *overdamped* Langevin equation is a first-order SDE,

$$dX_u = -\frac{1}{\gamma} V'(X_u) du + \sqrt{2D} dB_u, \quad D = \frac{k_B T}{\gamma}, \quad (7)$$

the last identity being Einstein’s relation between diffusivity, temperature and friction.

The density of X_u obeys a deterministic partial differential equation. For a general SDE $dX = b du + \sqrt{2D} dB$ it is the *Fokker–Planck* equation

$$\partial_u p = -\nabla \cdot (bp) + D \nabla^2 p, \quad (8)$$

which one obtains by applying [Section 1.2](#) to a test function and integrating by parts. For the gradient drift $b = -\nabla V/\gamma$ it can be written as a continuity equation with a current that vanishes when $p \propto e^{-V/(\gamma D)}$, so the stationary law is the *Boltzmann distribution*

$$p_\infty(x) \propto \exp\left[-\frac{V(x)}{k_B T}\right]. \quad (9)$$

(For the full second-order equation (6) the stationary law is the Maxwell–Boltzmann density in position *and* velocity, $\propto e^{-(V+\frac{1}{2}mv^2)/k_B T}$; its position marginal is again (9).)

Two consequences matter here. First, (9) closes a loop between dynamics and equilibrium: simulating (7) long enough samples the Boltzmann law, which is the basis of Langevin Monte Carlo. Second, the drift needed to do so is exactly a score,

$$\nabla \log p_\infty = -\frac{\nabla V}{k_B T}, \quad (10)$$

so *knowing a score is the same as knowing how to sample*. Generative diffusion is this observation applied to a family of distributions instead of one.

1.4 The Ornstein–Uhlenbeck process

Take the simplest confining potential, $V(x) = \frac{1}{2}\|x\|^2$. Then (7) becomes the *Ornstein–Uhlenbeck* process (Uhlenbeck and Ornstein, 1930),

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad (11)$$

in the normalisation used throughout this note. It is the unique process that is simultaneously Gaussian, Markov and stationary in continuous time, it is exactly solvable (Section 2.4), and its stationary law is the standard normal.

That last property is why diffusion models use it. The forward process needs a fixed, parameter-free distribution to end at, so that reverse sampling can be started from something known; $\mathcal{N}(0, I)$ is that distribution. The particular constants in (11) give the *variance-preserving* convention (Song et al., 2021): signal and noise scales satisfy $m_t^2 + \Delta_t = 1$, so data of unit variance keep unit variance at every t .

1.5 Diffusion models as non-equilibrium thermodynamics

Sohl-Dickstein et al. (2015) introduced the construction now called a diffusion model by explicit analogy with non-equilibrium thermodynamics. Their forward process is a chain of small perturbations that gradually converts a structured distribution into a simple one — an annealing schedule — and the generative model is a learned reversal of that chain. The training objective they derive is a variational bound of the same algebraic form as the work identities of non-equilibrium statistical mechanics: the log-likelihood is bounded by an average of log-ratios along the forward path, exactly as free-energy differences are bounded in Jarzynski-type equalities and estimated in annealed importance sampling. The analogy is structural rather than decorative — it is what supplies the objective — and it is the historical reason the field speaks of temperature, annealing and entropy at all.

The direction of the forward process is fixed by the second law.

Toolbox 4. The second law for the Fokker–Planck equation

Let p_t evolve by (8) with gradient drift and stationary law $p_\infty \propto e^{-V}$ (set $D = 1$, $\gamma = 1$, $k_B T = 1$). The relative entropy

$$\mathcal{F}[p_t] = D_{\text{KL}}(p_t \parallel p_\infty) = \int p_t \log \frac{p_t}{p_\infty} \quad (12)$$

is a free energy: it is the mean energy minus the entropy, up to a constant. Differentiate, use (8) written as $\partial_t p = \nabla \cdot (p \nabla \log(p/p_\infty))$, and integrate by parts:

$$\frac{d\mathcal{F}}{dt} = \int \partial_t p \log \frac{p}{p_\infty} = - \int p \left\| \nabla \log \frac{p}{p_\infty} \right\|^2 \leq 0. \quad (13)$$

The free energy decreases monotonically and the relative entropy is consumed at a rate given by a relative Fisher information; equality holds only at $p_t = p_\infty$. This is the H-theorem, and it is the second law for this dynamics. $\square$

Section 1.5 says the forward channel destroys information about the data monotonically and irreversibly, and

(13) identifies the rate. The reverse direction is not forbidden, but it is not free: it requires the score of the intermediate laws to be supplied from outside. [Anderson \(1982\)](#) makes this precise — the time-reversed process is again a diffusion, with a drift corrected by $\nabla \log p_t$. The entire modelling effort of a diffusion model is the estimation of that one object, and the entire content of this note is what that object looks like when the data are trajectories.

1.6 Two more tools from statistical mechanics

1.6.1 Fluctuation–dissipation, and linear response

Couple a system to a small external field h conjugate to an observable a , so that the Gibbs weight acquires a factor $e^{h \cdot a}$. The partition function $Z(h) = \int v(a) e^{h \cdot a} da$ is then a cumulant generating function, and

$$\frac{\partial \langle a_k \rangle}{\partial h_j} = \frac{\partial^2 \log Z}{\partial h_j \partial h_k} = \text{Cov}(a_k, a_j). \quad (14)$$

The response to a small perturbation is a spontaneous fluctuation. In equilibrium thermodynamics this is the fluctuation–dissipation relation — the magnetic susceptibility equals β times the magnetisation variance, the heat capacity equals β^2 times the energy variance ([Landau and Lifshitz, 1980](#)). Susceptibilities defined this way are the standard probe of phase transitions, and they are the tool used to detect the transitions in the training of energy-based models discussed below ([Bachtis et al., 2024](#)).

[Equation \(14\)](#) is a *static* identity between a derivative and a covariance in one fixed measure. [Section 2.7](#) shows that the response of the diffusion score to the observed data is exactly of this form. It is not a two-time dynamical fluctuation–dissipation theorem, and this note does not claim one.

1.6.2 First- and second-order phase transitions

A phase transition is a non-analyticity of the free energy in the thermodynamic limit, and its *order* is the order of the derivative that first misbehaves.

- At a **first-order** transition the first derivative of the free energy jumps. The order parameter is discontinuous, latent heat is absorbed, the two phases coexist at the transition, and correlation lengths stay finite.
- At a **second-order** transition the first derivative is continuous and the second diverges. The order parameter grows continuously from zero, the susceptibility (14) and the correlation length diverge, and observables acquire power laws.

Both appear in the statistical physics of generative models. The *speciation* transition of diffusion models — the diffusion time at which the reverse process commits to one class of the data rather than another — is analysed as a symmetry-breaking transition of this kind ([Biroli et al., 2024](#); [Achilli et al., 2026](#)), and the training of energy-based models has been described as a *cascade* of second-order transitions, one per direction that the model learns, each detected by the growth of the corresponding susceptibility ([Bachtis et al., 2024](#)). A related hierarchical picture is reported for images ([Sclocchi et al., 2025](#)).

No transition is claimed here

The model of this note has $T = 20$ frames and no thermodynamic limit is taken. Correlation lengths that grow until they reach the length of the chain and then stop are showing a finite-size ceiling, not a divergence. [Section 3](#) builds the diagnostics so that this distinction is visible rather than hidden, and [Section 10](#) reports where the ceiling is reached.

2 The object of study, and the identities it needs

2.1 One trajectory is one data point

We observe a planar trajectory at internal times

$$u_k = kh, \quad k = 0, 1, \dots, T-1.$$

The clean frame is $A_k \in \mathbb{R}^2$ and the complete datum is the stacked vector

$$A = (A_0, A_1, \dots, A_{T-1}) \in \mathbb{R}^{2T}. \quad (15)$$

One data point is one trajectory; the T frames are correlated coordinates of that single point, not T independent samples.

A second clock, the diffusion time $t \geq 0$, controls how strongly the whole vector is corrupted. The two clocks have different roles: internal time u_k describes the clean physical dynamics, diffusion time t describes the artificial noising channel of the generative model. [Figure 1](#) shows them acting on one trajectory.

Two clocks: internal time orders the frames, diffusion time corrupts them all at once

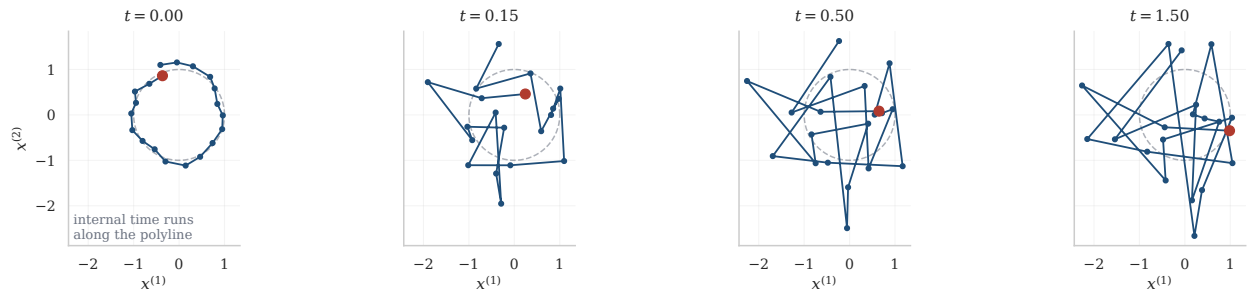


Figure 1. The two clocks. Internal time is the ordering of the points along the polyline. Diffusion time t shrinks and corrupts every frame at once. The same standardised noise realisation is used in all four panels, so the only difference between them is t .

2.2 Research question

Research question

How can the Markov structure of the clean rotating trajectory be exploited to compute and understand the dynamics of the joint diffusion score?

Four measurable subquestions make this operational.

1. How different is the joint score from a collection of one-frame marginal scores?
2. How strongly does the observed frame j influence the score acting on frame k ?
3. How does that influence decay with the temporal lag $|j - k|$?
4. How many neighbouring frames are needed to reproduce the full score to a given accuracy?

2.3 Notation

Symbol	Meaning
k, j	discrete internal-time indices
h	internal-time spacing between consecutive frames
t	diffusion time
A_k	clean Cartesian frame; A the stacked clean trajectory
X_k	noised Cartesian frame at diffusion time t
R_k, Θ_k	latent radius and angle of the polar model
$m_t = e^{-t}$	signal attenuation of the variance-preserving OU channel
$\Delta_t = 1 - e^{-2t}$	variance added by that channel
$S_k(x, t)$	k th two-dimensional block of the joint score
$J_{k,j}(x, t)$	block response $\partial S_k / \partial x_j \in \mathbb{R}^{2 \times 2}$
$\ell = j - k $	temporal lag
L	radius of a temporal observation window around frame k

2.4 The forward channel, solved

The surrogate and the polar process have different clean dynamics but the same noising channel, so we solve it once. The forward process in diffusion time is (11) applied to the whole trajectory,

$$dX_t = -X_t dt + \sqrt{2} dW_t, \quad X_0 = A, \quad (16)$$

with W_t a Brownian motion in $\mathbb{R}^{2T}$.

Toolbox 5. Solving the variance-preserving OU channel

Step 1: choose the integrating factor. Consider $\phi(x, t) = e^t x$. It is linear in x , so by (5) in Section 1.2 no Itô correction arises:

$$d(e^t X_t) = e^t X_t dt + e^t dX_t.$$

Substituting (16) the two drift terms cancel exactly,

$$d(e^t X_t) = e^t X_t dt + e^t (-X_t dt + \sqrt{2} dW_t) = \sqrt{2} e^t dW_t. \quad (17)$$

This is the whole purpose of the factor e^t : it removes the drift and leaves a pure stochastic differential, which can simply be integrated.

Step 2: integrate from 0 to t . The left side telescopes because it is an exact differential:

$$e^t X_t - e^0 X_0 = \sqrt{2} \int_0^t e^s dW_s \quad \implies \quad X_t = e^{-t} A + \sqrt{2} \int_0^t e^{-(t-s)} dW_s,$$

where e^{-t} has been taken inside the integral, which is legitimate because it does not depend on the integration variable s .

Step 3: identify the law of the noise term. The integrand $e^{-(t-s)}$ is deterministic, so by [Section 1.2](#) the integral is Gaussian, has mean zero by (2), and by the Itô isometry (3) has covariance

$$2 \int_0^t e^{-2(t-s)} ds \stackrel{v=t-s}{=} 2 \int_0^t e^{-2v} dv = 2 \cdot \frac{1 - e^{-2t}}{2} = 1 - e^{-2t},$$

in each of the $2T$ coordinates independently.

Step 4: collect.

$$X_t = m_t A + \sqrt{\Delta_t} Z, \quad m_t = e^{-t}, \quad \Delta_t = 1 - e^{-2t}, \quad Z \sim \mathcal{N}(0, I_{2T}). \quad (18)$$

Note $m_t^2 + \Delta_t = 1$: the channel is variance preserving, as promised in [Section 1.4](#). At $t = 0$ it is the identity; as $t \rightarrow \infty$ it forgets A entirely and returns $\mathcal{N}(0, I_{2T})$. $\square$

2.5 The joint score is a posterior denoiser

The noised density is the Gaussian smoothing of the data density,

$$p_t(x) = \int p_0(a) \mathcal{N}(x; m_t a, \Delta_t I_{2T}) da. \quad (19)$$

Differentiating under the integral, and using $\nabla_x \mathcal{N}(x; m_t a, \Delta_t I) = \mathcal{N}(x; m_t a, \Delta_t I) (m_t a - x) / \Delta_t$,

$$\nabla_x p_t(x) = \int p_0(a) \mathcal{N}(x; m_t a, \Delta_t I) \frac{m_t a - x}{\Delta_t} da. \quad (20)$$

Dividing by $p_t(x)$ turns the normalised integrand into the posterior $p(a | x)$ of the clean trajectory given the noisy one:

Score–posterior identity

$$S(x, t) = \nabla_x \log p_t(x) = \frac{m_t \mathbb{E}[A | X_t = x] - x}{\Delta_t} \quad (21)$$

and block by block,

$$S_k(x, t) = \frac{m_t \mathbb{E}[A_k | X_0 = x_0, \dots, X_{T-1} = x_{T-1}] - x_k}{\Delta_t}. \quad (22)$$

The score at frame k is therefore a *smoothing* estimate: it uses every observation, past and future, to estimate one clean frame.

2.6 Why the marginal score is a different object

The one-frame marginal score denoises frame k using frame k only,

$$s_k^{\text{marg}}(x_k, t) = \frac{m_t \mathbb{E}[A_k | X_k = x_k] - x_k}{\Delta_t}, \quad (23)$$

and subtracting it from (22) isolates the difference exactly:

$$S_k(x, t) - s_k^{\text{marg}}(x_k, t) = \frac{m_t}{\Delta_t} \left(\mathbb{E}[A_k | X_{0:T-1} = x] - \mathbb{E}[A_k | X_k = x_k] \right). \quad (24)$$

The gap is precisely the extra denoising information supplied by the rest of the trajectory. Figure 2 is the same statement as a picture: the joint score is a smoother, in which every observation can inform every clean frame, while the product-of-marginals score deletes all cross-frame information before denoising begins.

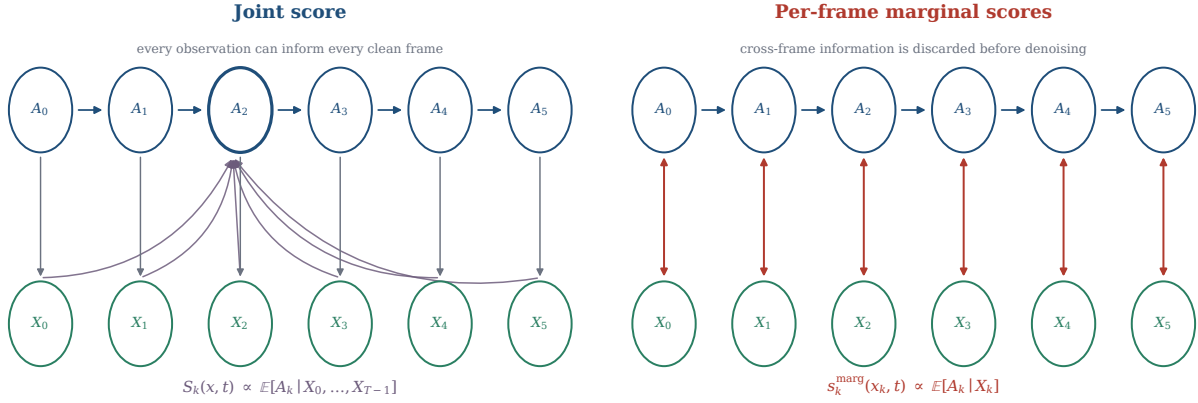


Figure 2. The two objects compared in (24), as graphical models. Clean frames A_k (blue) form a Markov chain in internal time and each emits one noisy observation X_k (green) through the channel (18). Left: computing S_2 requires the posterior of A_2 given *all* observations, so every X_j has a path to A_2 . Right: the per-frame marginal score conditions on X_k alone, which severs every cross-frame path. The size of the difference is measured in Sections 6 and 10.

2.7 The response of the score is a posterior covariance

Question 2 of the research question asks how strongly the observation at frame j influences the score at frame k . The answer is an identity, and it is worth deriving slowly because everything measured later is an instance of it.

Toolbox 6. From the observation as an external field to the response identity

Step 1: write the posterior as a Gibbs measure with an external field. By Bayes and (18),

$$\begin{aligned} p(a | x) &\propto p_0(a) \exp \left[-\frac{\|x - m_t a\|^2}{2\Delta_t} \right] \\ &= p_0(a) \underbrace{\exp \left[-\frac{\|x\|^2}{2\Delta_t} \right]}_{\text{independent of } a} \exp \left[-\frac{m_t^2}{2\Delta_t} \|a\|^2 + \frac{m_t}{\Delta_t} x^\top a \right] \\ &\propto \underbrace{p_0(a) \exp \left[-\frac{m_t^2}{2\Delta_t} \|a\|^2 \right]}_{\nu(a), \text{ independent of } x} e^{h^\top a}, \quad \boxed{h = \frac{m_t}{\Delta_t} x}. \end{aligned} \quad (25)$$

The expansion of $\|x - m_t a\|^2$ has produced exactly three kinds of term: one that does not involve a and is absorbed by the normalisation, one quadratic in a that does not involve x , and one bilinear term $x^\top a$. The whole dependence on the data therefore enters through a single vector h , coupled *linearly* to A . This is the structure of Section 1.6: the observation at frame j acts as an external field $h_j = (m_t/\Delta_t)x_j$

conjugate to the clean state A_j .

Step 2: differentiate the log-normalisation once. Write $Z(h) = \int v(a) e^{h^\top a} da$, so that $p(a | x) = v(a) e^{h^\top a} / Z(h)$. Then

$$\frac{\partial \log Z}{\partial h_j} = \frac{1}{Z} \int a_j v(a) e^{h^\top a} da = \mathbb{E}[A_j | x]. \quad (26)$$

$\log Z$ is the cumulant generating function of the posterior.

Step 3: differentiate twice. Differentiating (26) again, and using $\partial_{h_k} e^{h^\top a} = a_k e^{h^\top a}$ together with $\partial_{h_k} Z^{-1} = -Z^{-2} \partial_{h_k} Z$,

$$\frac{\partial^2 \log Z}{\partial h_k \partial h_j} = \mathbb{E}[A_k A_j | x] - \mathbb{E}[A_k | x] \mathbb{E}[A_j | x] = \text{Cov}(A_k, A_j | x). \quad (27)$$

Combining the last two displays, the derivative of a posterior mean with respect to the field is a posterior covariance:

$$\frac{\partial}{\partial h_j} \mathbb{E}[A_k | x] = \text{Cov}(A_k, A_j | x). \quad (28)$$

Step 4: change variables from the field back to the data. Since $h = (m_t / \Delta_t) x$ the chain rule gives $\partial / \partial x_j = (m_t / \Delta_t) \partial / \partial h_j$, hence

$$\boxed{\frac{\partial}{\partial x_j} \mathbb{E}[A_k | x] = \frac{m_t}{\Delta_t} \text{Cov}(A_k, A_j | x).} \quad (29)$$

Step 5: differentiate the score. Apply $\partial / \partial x_j$ to $S_k = (m_t \mathbb{E}[A_k | x] - x_k) / \Delta_t$ from (22). The first term contributes (m_t / Δ_t) times (29); the second contributes $-\delta_{kj} I_2 / \Delta_t$. Therefore

$$\boxed{J_{kj}(x, t) := \frac{\partial S_k}{\partial x_j} = -\frac{\delta_{kj}}{\Delta_t} I_2 + \frac{m_t^2}{\Delta_t^2} \text{Cov}(A_k, A_j | X_t = x).} \quad (30)$$

Here $\text{Cov}(A_k, A_j | x)$ is the 2×2 cross-covariance block between the two clean frames under the posterior. $\square$

Consistency check. J is the Hessian $\nabla^2 \log p_t$, so it must be symmetric under exchanging the pairs (k, α) and (j, β) . It is: transposing the 2×2 block of (30) exchanges A_k and A_j , and $\delta_{kj} I_2$ is symmetric.

What the response identity says

For $j \neq k$ the response is proportional to a posterior cross-covariance. Perturbing a distant observed frame changes S_k if and only if the two clean states remain statistically coupled *after* conditioning on the whole noisy trajectory. The pattern of the score Jacobian is therefore a direct map of posterior information flow, and the prefactor m_t^2 / Δ_t^2 — which tends to zero as t grows — is separate from that pattern. Keeping the prefactor and the pattern apart is the point of [Section 3](#).

3 Diagnostics: what to measure, and why

Subquestions 2–4 ask about strength, reach, and sufficiency. These are three different properties of one profile, and in the measurements below they move in different directions. This section defines them, then builds one combined diagnostic, and states what each cannot see.

Throughout, fix a central frame k and let

$$C_t(\ell) = \mathbb{E}_x [\|J_{k,k+\ell}(x, t)\|_F], \quad \ell = 0, 1, \dots, L_{\max}, \quad (31)$$

be the *response profile*: the average Frobenius norm of the 2×2 response block at lag ℓ , averaged over noisy trajectories and over the two sides $k \pm \ell$. The term $C_t(0)$ is the self-response, and by (30) it is dominated by $1/\Delta_t$.

3.1 Strength: response intensity

Intensity

$$I_{\text{off}}(t) = \sum_{\ell \geq 1} C_t(\ell) \quad (32)$$

This is the total amount of cross-frame response, in the units of the score Jacobian. It answers “how much do the other frames matter at all”, and it says nothing about where they sit.

3.2 Reach: three candidate lengths

The natural first choice is an e-folding length: when the profile is close to exponential,

$$C_t(\ell) \approx C_t(1) \exp[-(\ell - 1)/\xi^{\text{exp}}],$$

and ξ^{exp} is obtained by least squares on $\log C_t$. It is the quantity a statistical-mechanics reading of the problem would want, because it is the finite-chain estimate of a correlation length. It is also fragile: once the measured profile is nearly flat the fit is ill-conditioned, and it becomes unbounded. In our measurements the fitted $\xi_{\theta}^{\text{exp}}$ exceeds 290 frames in a chain of 20 at the largest diffusion times, which is not a length but a fit artefact. We therefore report it only in the low-noise regime where it is meaningful (Table 3).

A bounded alternative is the normalised range used in the first version of this study,

$$\xi^{\text{norm}}(t) = \sum_{\ell=1}^{L_{\max}} \frac{C_t(\ell)}{C_t(1)}, \quad (33)$$

which is always between 1 and $L_{\max}$. It is stable, but it is a pure *shape* statistic: multiplying the whole profile by a constant leaves it unchanged. It therefore cannot distinguish a strong short-range coupling from a negligible long-range one.

The third choice, and the one used from here on, is the first moment of the response mass:

Weighted mean lag

$$\bar{\ell}(t) = \frac{\sum_{\ell \geq 1} \ell C_t(\ell)}{\sum_{\ell \geq 1} C_t(\ell)} = \frac{1}{I_{\text{off}}(t)} \sum_{\ell \geq 1} \ell C_t(\ell) \quad (34)$$

$\bar{\ell}$ is the centre of mass of the cross-frame response: the average distance, in frames, from which the response actually arrives. It is measured in frames, it is bounded by $L_{\max}$, and — unlike ξ^{exp} — it does not assume the profile has any particular shape. It has a known *structureless ceiling*: a completely flat profile gives

$\bar{\ell} = (1 + L_{\max})/2$, which for our $L_{\max} = 10$ is 5.5. A measured $\bar{\ell}$ close to 5.5 means that the temporal structure has been erased, not that the correlation is long.

3.3 Both at once: a weighted reach

Intensity and mean lag answer complementary questions, and the product of the two is a single number that responds to either.

Weighted reach, absolute and relative

$$\Xi(t) = I_{\text{off}}(t) \bar{\ell}(t) = \sum_{\ell \geq 1} \ell C_t(\ell), \quad \tilde{\Xi}(t) = \frac{\Xi(t)}{C_t(0)} \quad (35)$$

Three remarks on the construction.

1. Ξ is the unnormalised first moment of the profile. It increases if the coupling becomes stronger *or* if it arrives from farther away, and it factorises exactly into “how much” times “how far”. That factorisation is the reason to prefer it to an ad hoc combination: the two ingredients can always be read off separately.
2. Dividing by the self-response $C_t(0)$ makes it comparable across diffusion times. By (30), $C_t(0) \simeq 1/\Delta_t$ up to the posterior variance term, so $\tilde{\Xi}$ removes the trivial overall scale of the Jacobian and measures the reach of the context *relative to the local term* that competes with it. It is still measured in frames.
3. $\tilde{\Xi}$ inherits a ceiling from $\bar{\ell}$, so it too must be read against the structureless value rather than in absolute terms.

Figure 3(a,b) is the construction test. Four reference profiles are built by crossing two shapes (short, long) with two amplitudes (strong, weak). By design ξ^{norm} cannot separate strong from weak, I_{off} cannot separate short from long, $\bar{\ell}$ again cannot separate strong from weak — and Ξ orders all four. Panel (c) shows why this matters in the real measurement, and is discussed in Section 10.

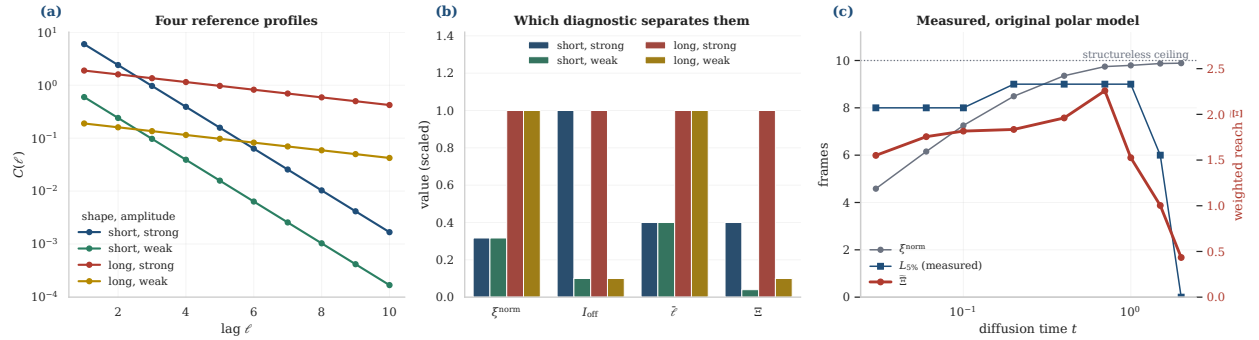


Figure 3. Building a diagnostic that sees reach and strength together. (a) Four synthetic reference profiles: two decay lengths crossed with two amplitudes. (b) The four diagnostics evaluated on them, each scaled by its own maximum. ξ^{norm} and $\bar{\ell}$ are blind to amplitude, I_{off} is blind to reach, and only $\Xi = I_{\text{off}}\bar{\ell}$ separates all four. (c) Measured in the polar model of Section 7: the normalised range rises monotonically to its structureless ceiling, whereas the weighted reach $\tilde{\Xi}$ and the independently measured window radius $L_{5\%}$ both rise and then fall.

3.4 Sufficiency: the functional receptive field

Every diagnostic so far is infinitesimal — it is built from derivatives. A more direct test of subquestion 4 is to recompute the score from a truncated observation window $x_{k-L:k+L}$, ignoring the rest of the trajectory

entirely. Writing $S_k^{(L)}$ for that windowed score,

Window error and receptive radius

$$\varepsilon_L(t) = \left[\frac{\mathbb{E} \|S_k^{(L)} - S_k^{\text{full}}\|^2}{\mathbb{E} \|S_k^{\text{full}}\|^2} \right]^{1/2}, \quad L_\epsilon(t) = \min\{L : \varepsilon_L(t) \leq \epsilon\}. \quad (36)$$

L_ϵ is the diagnostic with the most direct modelling meaning: it is the temporal receptive field a network would need in order to represent the score to accuracy ϵ at that noise level. It is also the most expensive to compute, which is why the infinitesimal diagnostics are worth having.

What each diagnostic cannot see

A large reach need not mean a large effect. If every off-diagonal response is tiny but nearly equal, then ξ^{norm} and $\bar{\ell}$ both saturate at their structureless ceilings while the total correction to the score is negligible. This is exactly what happens at large diffusion time, where the score approaches the universal field $-x$. Conversely I_{off} alone cannot tell a tightly local coupling from a diffuse one. Only the pair, or the weighted reach that combines them, is safe to quote.

Part I

An exactly solvable surrogate

4 The surrogate, and the rotation as a gauge

4.1 The model

The surrogate keeps a ring, a coherent rotation, and statistical dependence along the trajectory, and drops everything else:

$$Z_0 \sim p_\lambda(z) \propto \exp \left[-\frac{(\|z\| - 1)^2}{2\lambda} \right], \quad (37)$$

$$Z_{k+1} = R_\psi Z_k + \sigma \eta_{k+1}, \quad \eta_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2), \quad (38)$$

with R_ψ the planar rotation through the one-frame angle ψ .

The surrogate is not a discretisation of the polar model

In (38) the ring acts only on Z_0 . Later frames form a rotating Cartesian random walk and their radial variance grows without bound. In the polar process of [Section 7](#) a confining force acts at *every* internal time and continually returns the radius to one. The surrogate is a solvable companion, and its role is to make the mechanism visible in closed form, not to approximate the board model.

4.2 The rotation matrix

Because the rotation is the object we are about to remove, it is worth writing out.

$$R_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}, \quad R_\psi \begin{pmatrix} z^{(1)} \\ z^{(2)} \end{pmatrix} = \begin{pmatrix} z^{(1)} \cos \psi - z^{(2)} \sin \psi \\ z^{(1)} \sin \psi + z^{(2)} \cos \psi \end{pmatrix}. \quad (39)$$

Its columns are $R_\psi e_1 = (\cos \psi, \sin \psi)$ and $R_\psi e_2 = (-\sin \psi, \cos \psi)$: the two standard basis vectors, each turned through ψ . They are unit vectors, since $\cos^2 \psi + \sin^2 \psi = 1$, and they are orthogonal, since $\cos \psi (-\sin \psi) + \sin \psi \cos \psi = 0$. In matrix form,

$$R_\psi^\top R_\psi = I_2, \quad \det R_\psi = \cos^2 \psi + \sin^2 \psi = 1, \quad R_\psi^\top = R_{-\psi}, \quad R_\psi R_\varphi = R_{\psi+\varphi}. \quad (40)$$

Orthogonality is what we use: it means R_ψ preserves inner products and hence lengths,

$$\|R_\psi z\|^2 = (R_\psi z)^\top (R_\psi z) = z^\top R_\psi^\top R_\psi z = z^\top z = \|z\|^2. \quad (41)$$

[Figure 4](#) shows the three facts side by side.

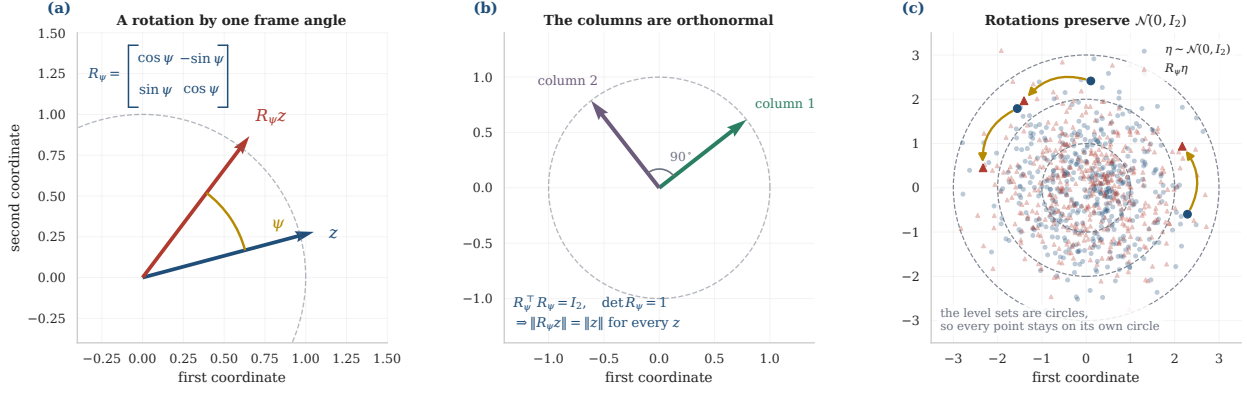


Figure 4. The rotation matrix. (a) R_ψ turns every vector through the same angle ψ without changing its length. (b) Its columns are the rotated standard basis: unit vectors at right angles, which is the statement $R_\psi^T R_\psi = I_2$. (c) The standard isotropic Gaussian is unchanged, because its level sets are circles centred at the origin and a rotation maps each circle to itself; three sample points and their images are highlighted.

4.3 Rotations preserve the standard isotropic Gaussian

This small lemma is used twice below, so we prove it twice — once by covariance and once by density, because the two arguments illuminate different things.

By covariance. Let $\eta \sim \mathcal{N}(0, I_2)$ and $\tilde{\eta} = R\eta$ with R orthogonal. A linear map of a Gaussian is Gaussian, so $\tilde{\eta}$ is determined by its first two moments:

$$\mathbb{E}[\tilde{\eta}] = R \mathbb{E}[\eta] = 0, \quad \text{Cov}(\tilde{\eta}) = R \text{Cov}(\eta) R^T = R I_2 R^T = R R^T = I_2. \quad (42)$$

Hence $\tilde{\eta} \sim \mathcal{N}(0, I_2)$: the same law, not merely a similar one.

By density. The density $p(z) = (2\pi)^{-1} e^{-\|z\|^2/2}$ depends on z only through $\|z\|$. Under $z \mapsto Rz$ the change-of-variables formula gives the density $p(R^{-1}z) |\det R^{-1}|$, and both factors are unchanged: $\|R^{-1}z\| = \|z\|$ by (41), and $|\det R^{-1}| = 1$ by (40). This is the version drawn in Figure 4(c): the level sets are circles, and a rotation slides each circle along itself.

The isotropy is essential. A rotation would *not* preserve $\mathcal{N}(0, \Sigma)$ for general Σ , and it is only because the innovation in (38) and the channel (18) are both isotropic that the reduction below is exact.

4.4 The gauge transformation

De-rotation

$$Y_k = R_{-k\psi} Z_k \quad (43)$$

Y_k is the trajectory as seen by an observer who turns with it: frame k is rotated *back* by the total angle $k\psi$ accumulated up to that frame. Substituting (38) and using the composition rule (40),

$$\begin{aligned} Y_{k+1} &= R_{-(k+1)\psi} (R_\psi Z_k + \sigma \eta_{k+1}) \\ &= \underbrace{R_{-(k+1)\psi} R_\psi}_{=R_{-k\psi}} Z_k + \sigma R_{-(k+1)\psi} \eta_{k+1} \\ &= Y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} := R_{-(k+1)\psi} \eta_{k+1}. \end{aligned} \quad (44)$$

By [Section 4.3](#) each $\tilde{\eta}_{k+1} \sim \mathcal{N}(0, I_2)$, and they remain independent because each is a fixed linear map of a different independent η . The rotation has vanished:

Gauge-reduced surrogate

$$Y_{k+1} = Y_k + \sigma \tilde{\eta}_{k+1}, \quad \tilde{\eta}_{k+1} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_2). \quad (45)$$

Why “gauge”. The term is borrowed from physics, where a gauge transformation is a change of description that leaves the physics untouched and can be chosen to simplify the equations. Here (43) is a known, exactly invertible, orthogonal change of coordinates. It removes no information and introduces none: the two descriptions are the same problem in different frames, and we are free to work in whichever is easier. That is the sense in which the rotation “costs no statistical difficulty”. It would not be true of an *unknown* ψ , which is a genuine latent variable — see [Section 13](#).

Stacking the per-frame rotations into one $2T \times 2T$ matrix,

$$U_\psi = \text{diag}(I_2, R_{-\psi}, R_{-2\psi}, \dots, R_{-(T-1)\psi}), \quad y = U_\psi z. \quad (46)$$

U_ψ is orthogonal, and how the score transforms

U_ψ is block diagonal with orthogonal blocks, so

$$U_\psi^\top U_\psi = \text{diag}(I_2^\top I_2, R_{-\psi}^\top R_{-\psi}, \dots) = \text{diag}(I_2, \dots, I_2) = I_{2T}, \quad \det U_\psi = \prod_k \det R_{-k\psi} = 1. \quad (47)$$

Consequently, writing $p_t^{(0)}$ and $S^{(0)}$ for the $\psi = 0$ quantities,

$$p_t(z | \psi) = p_t^{(0)}(U_\psi z), \quad S_\psi(z, t) = U_\psi^\top S^{(0)}(U_\psi z, t). \quad (48)$$

Both halves of (48) deserve a line of proof, since they are the reason the whole of Part I is computable.

The density. Apply U_ψ to the channel (18): $U_\psi X = m_t U_\psi A + \sqrt{\Delta_t} U_\psi Z$, and $U_\psi Z \sim \mathcal{N}(0, U_\psi U_\psi^\top) = \mathcal{N}(0, I)$ by (47) and [Section 4.3](#). So $U_\psi X$ has exactly the law of the noised $\psi = 0$ trajectory — the de-rotation commutes with the channel, which is the content of [Figure 5](#). For the densities, $Z = U_\psi^\top Y$ gives $p_t(z | \psi) = p_t^{(0)}(U_\psi z) |\det U_\psi| = p_t^{(0)}(U_\psi z)$, the Jacobian factor being 1.

The score. Differentiate the first identity with the chain rule. In components, with $y = U_\psi z$, so that $\partial y_b / \partial z_a = (U_\psi)_{ba}$,

$$\frac{\partial}{\partial z_a} \log p_t^{(0)}(U_\psi z) = \sum_b (U_\psi)_{ba} (\nabla \log p_t^{(0)})_b(U_\psi z) = [U_\psi^\top S^{(0)}(U_\psi z, t)]_a. \quad (49)$$

The transpose is not cosmetic: a score is a gradient, so it transforms with $U_\psi^\top$, while a point transforms with U_ψ . Since $U_\psi^\top = U_\psi^{-1}$ here, the recipe is simply: *de-rotate the data, solve the non-rotating problem, rotate the score back*.

U_ψ is orthogonal, so the de-rotation commutes with the isotropic channel

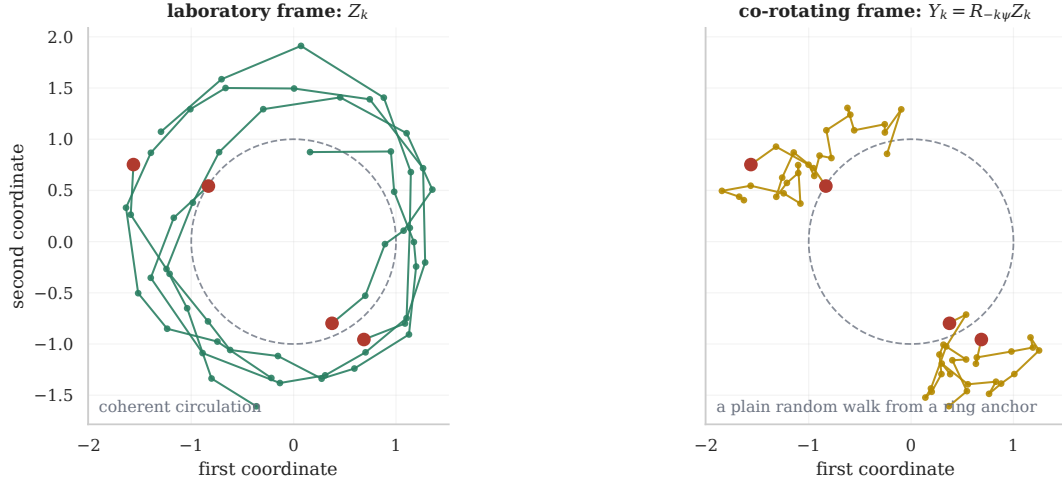


Figure 5. The gauge, on four trajectories ($T = 14$, $\sigma = 0.16$, $\psi = 2\pi/14$; large dots mark $k = 0$). Left: the data, coherently circulating. Right: the same trajectories after U_ψ . The circulation is gone and a random walk anchored on the ring remains. Because U_ψ is orthogonal it commutes with the isotropic channel, so the reduction is exact at every diffusion time; the round trip is verified to machine precision (Table 4).

4.5 Unrolling the recursion

Equation (45) is a random walk, and it is worth writing the first few steps explicitly because the resulting structure — one shared anchor plus accumulated noise — is what makes the model solvable.

$$\begin{aligned} Y_1 &= Y_0 + \sigma \tilde{\eta}_1, \\ Y_2 &= Y_1 + \sigma \tilde{\eta}_2 = Y_0 + \sigma (\tilde{\eta}_1 + \tilde{\eta}_2), \\ Y_3 &= Y_2 + \sigma \tilde{\eta}_3 = Y_0 + \sigma (\tilde{\eta}_1 + \tilde{\eta}_2 + \tilde{\eta}_3), \quad \text{and so on.} \end{aligned}$$

Formally, sum the increments and let the intermediate terms cancel:

$$Y_k - Y_0 = \sum_{r=1}^k (Y_r - Y_{r-1}) = \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad (50)$$

a telescoping sum in which every Y_r with $0 < r < k$ appears once with each sign. Hence

Shared anchor plus accumulated noise

$$Y_k = A + \sigma \sum_{r=1}^k \tilde{\eta}_r, \quad A := Y_0 \sim p_\lambda, \quad k = 0, \dots, T-1 \quad (51)$$

(the empty sum at $k = 0$ being zero).

Every frame contains the *same* random anchor A , drawn once on the ring, plus a partial sum of independent innovations. The anchor is a single two-dimensional variable shared by all T frames, and it is the only non-Gaussian ingredient in the model.

5 The clean law and the exact noised score

5.1 Clean joint density and clean score

From (45) the clean law in the co-rotating frame factorises over the ring prior and the one-step transitions,

$$p_0(y_0, \dots, y_{T-1}) = p_\lambda(y_0) \prod_{k=0}^{T-2} \mathcal{N}(y_{k+1}; y_k, \sigma^2 I_2), \quad (52)$$

so that

$$\log p_0(y) = -\frac{(\|y_0\| - 1)^2}{2\lambda} - \frac{1}{2\sigma^2} \sum_{k=0}^{T-2} \|y_{k+1} - y_k\|^2 + \text{const.} \quad (53)$$

Differentiating (53) is elementary but instructive, so we do it in full. For the ring term, use $\nabla_y \|y\| = y/\|y\|$:

$$\nabla_{y_0} \left[-\frac{(\|y_0\| - 1)^2}{2\lambda} \right] = -\frac{\|y_0\| - 1}{\lambda} \cdot \frac{y_0}{\|y_0\|} = \frac{1 - \|y_0\|}{\lambda} \cdot \frac{y_0}{\|y_0\|}, \quad (54)$$

which points inward when $\|y_0\| > 1$ and outward when $\|y_0\| < 1$.

For the transition terms, the key observation is a counting one: *an interior y_k appears in exactly two summands of (53)*, namely the edge $(k-1, k)$ that arrives at it and the edge $(k, k+1)$ that leaves it. Differentiating each,

$$\nabla_{y_k} \left[-\frac{1}{2\sigma^2} \|y_k - y_{k-1}\|^2 \right] = -\frac{y_k - y_{k-1}}{\sigma^2} \quad (\text{the incoming edge}), \quad (55)$$

$$\nabla_{y_k} \left[-\frac{1}{2\sigma^2} \|y_{k+1} - y_k\|^2 \right] = +\frac{y_{k+1} - y_k}{\sigma^2} \quad (\text{the outgoing edge}), \quad (56)$$

and adding them,

$$-\frac{y_k - y_{k-1}}{\sigma^2} + \frac{y_{k+1} - y_k}{\sigma^2} = \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}. \quad (57)$$

The boundary frames have only one incident edge each, and y_0 additionally carries (54). Collecting:

Clean score of the gauged surrogate

$$S_0(y, 0) = \frac{1 - \|y_0\|}{\lambda} \frac{y_0}{\|y_0\|} + \frac{y_1 - y_0}{\sigma^2}, \quad (58)$$

$$S_k(y, 0) = \frac{y_{k-1} - 2y_k + y_{k+1}}{\sigma^2}, \quad 1 \leq k \leq T-2, \quad (59)$$

$$S_{T-1}(y, 0) = \frac{y_{T-2} - y_{T-1}}{\sigma^2}. \quad (60)$$

Equation (59) is a discrete second derivative — a discrete Laplacian along internal time. Three consequences are worth stating, because they are what the noised score will destroy.

1. **It is strictly local.** S_k involves y_{k-1}, y_k, y_{k+1} and nothing else. Each interior block measures how far frame k deviates from the average of its two neighbours: it is a curvature, or an acceleration, along the path.
2. **It vanishes on the model's own solution.** A second difference is zero on any straight line in the co-

rotating frame. A trajectory that rotates perfectly at radius exactly 1 maps under (43) to a *constant* Y_k , for which every second difference vanishes and, since $\|y_0\| = 1$, the ring term (54) vanishes too. The clean score is then exactly zero. This is checked numerically and is exact to machine zero (Table 4).

3. **Only the first frame feels the ring.** The anchor enters (58) alone; the rest of the trajectory knows about the ring only indirectly, through the chain.



Figure 6. The clean surrogate score, on a path that co-rotates on the ring except for one displaced frame ($T = 7$, $\sigma = 0.15$, $\lambda = 0.05$). (a) The score blocks as arrows; lengths are square-root compressed, since the magnitudes span nearly two decades. (b) The true magnitudes: only the displaced frame and its two neighbours carry an appreciable score, an order of magnitude above the rest, and frame 0 additionally feels the ring anchor. (c) Why: an interior y_k occurs in exactly two summands of (53), one per incident edge, and the two derivatives (55)–(56) combine into a second difference. This locality does *not* survive noising, as Section 5.4 shows.

5.2 The noised law, in Kronecker notation

Condition on the anchor. From (51) the frames are Gaussian with

$$\mathbb{E}[Y_i | A = a] = a, \quad \text{Cov}(Y_i, Y_j | A = a) = \sigma^2 \mathbb{E}\left[\left(\sum_{r \leq i} \tilde{\eta}_r\right)\left(\sum_{s \leq j} \tilde{\eta}_s\right)^\top\right] = \sigma^2 \min(i, j) I_2, \quad (61)$$

because $\mathbb{E}[\tilde{\eta}_r \tilde{\eta}_s^\top] = \delta_{rs} I_2$, so only the $\min(i, j)$ indices common to both sums survive. Compactly,

$$Y | A = a \sim \mathcal{N}(\mathbf{1} \otimes a, K \otimes I_2), \quad K_{ij} = \sigma^2 \min(i, j), \quad i, j = 0, \dots, T-1. \quad (62)$$

Reading (62)

Y is the stacked $2T$ -vector $(Y_0^\top, \dots, Y_{T-1}^\top)^\top$, and $\mathbf{1} \in \mathbb{R}^T$ is the all-ones vector. The two Kronecker products separate the *temporal* index from the *spatial* one.

- $\mathbf{1} \otimes a = (a^\top, a^\top, \dots, a^\top)^\top \in \mathbb{R}^{2T}$ repeats the same two-dimensional anchor in every frame. This is (51): all frames have the same conditional mean.
- $K \otimes I_2$ has entries $(K \otimes I_2)_{(i, \alpha), (j, \beta)} = K_{ij} \delta_{\alpha\beta} = \text{Cov}(Y_i^{(\alpha)}, Y_j^{(\beta)} | A)$. So K carries all the temporal correlation, and I_2 says that the two spatial coordinates are uncorrelated and share the *same* temporal covariance — a consequence of the isotropy of the innovations.

For $T = 3$, written out with σ^2 factored out of K ,

$$K = \sigma^2 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad K \otimes I_2 = \sigma^2 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 2 \end{pmatrix}.$$

The zero first row and column say that $Y_0 = A$ has no innovation noise of its own: given the anchor it is known exactly. Each scalar entry of K has become a 2×2 diagonal block.

Now apply the channel. With $X = m_t Y + \sqrt{\Delta_t} \Xi$, $\Xi \sim \mathcal{N}(0, I_{2T})$ independent of Y , and using $I_{2T} = I_T \otimes I_2$ to combine the two covariances,

$$m_t^2 (K \otimes I_2) + \Delta_t (I_T \otimes I_2) = (m_t^2 K + \Delta_t I_T) \otimes I_2, \quad (63)$$

so that

$$X_t \mid A = a \sim \mathcal{N}(m_t \mathbf{1} \otimes a, C_t \otimes I_2), \quad \boxed{C_t = m_t^2 K + \Delta_t I_T}. \quad (64)$$

The crucial feature is that C_t does *not* depend on a : the noised law is a two-parameter *location* mixture of one fixed Gaussian.

5.3 The anchor posterior is two-dimensional

Define

$$q = C_t^{-1} \mathbf{1} \in \mathbb{R}^T, \quad \beta = m_t^2 \mathbf{1}^\top C_t^{-1} \mathbf{1} \in \mathbb{R}, \quad h = m_t \sum_{j=0}^{T-1} q_j x_j \in \mathbb{R}^2. \quad (65)$$

Expanding the exponent of (64) in powers of a , exactly as in Step 1 of Section 2.7 but now with the correlated covariance $C_t \otimes I_2$, gives

$$p(a \mid x) \propto \exp \left[-\frac{(\|a\| - 1)^2}{2\lambda} - \frac{\beta}{2} \|a\|^2 + h^\top a \right]. \quad (66)$$

The $2T$ -dimensional integral over clean trajectories has collapsed to a *two-dimensional* posterior over the shared anchor. That is the entire simplification, and it is a direct consequence of (51): conditionally on A everything else is Gaussian and can be integrated out in closed form.

Two moments of (66) are needed, $\mathbb{E}[A \mid x]$ and $\text{Cov}(A \mid x)$. Both are cheap. The prior and the $\|a\|^2$ term are rotationally symmetric, so the posterior depends on the direction of a only through $h^\top a$; therefore

$$\mathbb{E}[A \mid x] = \mu(\rho) \widehat{h}, \quad \text{Cov}(A \mid x) = v_\perp I_2 + (v_\parallel - v_\perp) \widehat{h} \widehat{h}^\top, \quad \rho = \|h\|, \quad \widehat{h} = h/\rho, \quad (67)$$

with three scalar functions of the single scalar ρ . They are tabulated once per $(t, \text{index set})$ by one-dimensional quadrature and interpolated.¹

¹Doing the angular integral analytically produces modified Bessel functions; Appendix A records the resulting one-dimensional integrals for reproducibility. Nothing in the argument depends on them — the posterior is two-dimensional, so any quadrature will

5.4 Exact score and exact response

Write $\log p_t(x) = \text{const} - \frac{1}{2}x^\top (C_t^{-1} \otimes I_2)x + \log Z(h(x))$, with Z the normalisation of (66). The gradient of the first term is $-(C_t^{-1}x)_k$ in block k ; for the second, $\partial h/\partial x_k = m_t q_k I_2$ and $\nabla_h \log Z = \mathbb{E}[A | x]$ by (26). Hence

Exact surrogate score

$$S_k(x, t) = -(C_t^{-1}x)_k + m_t q_k \mathbb{E}[A | x]. \quad (68)$$

The first term is the noised random-walk contribution, near-diagonal in k ; the second is a global correction carried by the one shared anchor. Differentiating once more, with $\partial_{x_j} \mathbb{E}[A | x] = m_t q_j \text{Cov}(A | x)$ from (28),

Exact surrogate response

$$J_{kj}(x, t) = -(C_t^{-1})_{kj} I_2 + m_t^2 q_k q_j \text{Cov}(A | x). \quad (69)$$

The temporal coefficient of the second term is the outer product $qq^\top$, which is **rank one**. This makes the origin of the long-range part of the response completely explicit: it is not a chain of local couplings but a single shared latent variable seen by every frame. Figure 7 separates the two terms.

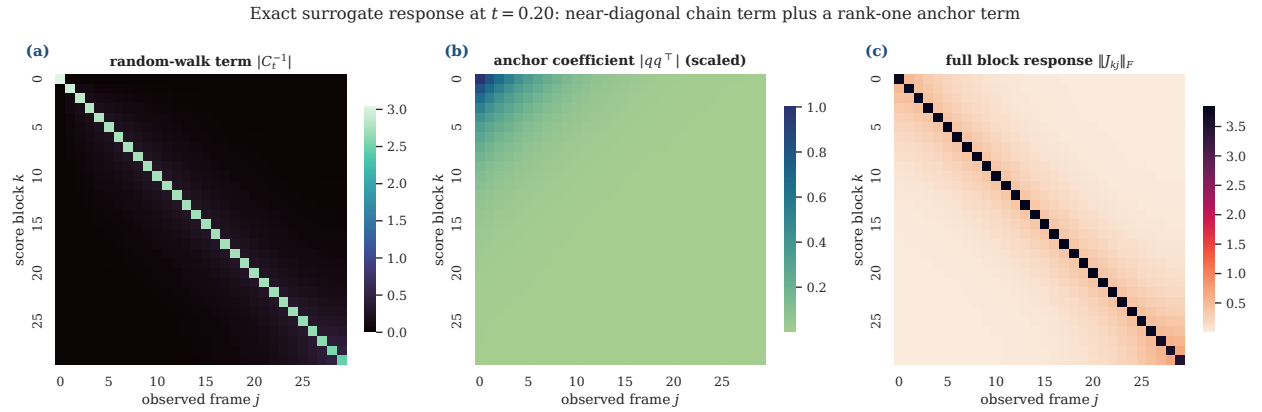


Figure 7. Exact decomposition of the surrogate response at $t = 0.20$, $T = 30$. (a) The random-walk term $|C_t^{-1}|$ is concentrated near the diagonal. (b) The anchor coefficient $|qq^\top|$ is rank one and spread over all lags. (c) The average norm of the complete 2×2 blocks. The analytic Jacobian agrees with central finite differences to 9.2×10^{-6} (Table 4).

What the surrogate settles

Clean Markov locality does not survive marginalisation through the OU channel as locality of the noised score: (69) has nonzero blocks at every lag for every $t > 0$. But the nonlocal part is not arbitrary. It decomposes exactly into a near-diagonal chain term and a rank-one correction generated by the shared ring anchor.

do — and they are not used again in the main text.

6 What the surrogate measures

Protocol. $T = 30$, $\lambda = 0.05$, $\sigma = 0.15$, $\psi = 2\pi/30$; 350 clean trajectories, corrupted independently at each diffusion time; anchor moments on 180 quadrature nodes and a 600-point interpolation grid in $\|h\|$. In this part the response profile (31) is averaged over *all* frame pairs at equal lag rather than around one central frame, so $L_{\max} = T - 1 = 29$ and the structureless ceilings are $\bar{\ell} = 15$ and $\xi^{\text{norm}} = 29$. The window error (36) is measured at the central frame $k = 15$.

Table 1. Surrogate diagnostics. “Marginal RMSE” is the relative RMS error of replacing the joint score by per-frame marginal scores. $L_{5\%}$ is the window radius of (36). The last row is the value a completely flat profile would give.

t	marginal RMSE	I_{off}	$\bar{\ell}$	ξ^{norm}	$\tilde{\Xi}$	$L_{5\%}$
0.05	0.959	6.093	2.73	2.73	1.467	6
0.20	0.779	2.177	5.34	5.32	3.067	10
0.70	0.462	1.171	10.61	12.61	6.994	13
1.50	0.154	0.703	13.30	20.68	6.430	11
<i>flat profile</i>	—	—	15.00	29.00	—	—

Exactly solvable surrogate: the four diagnostics

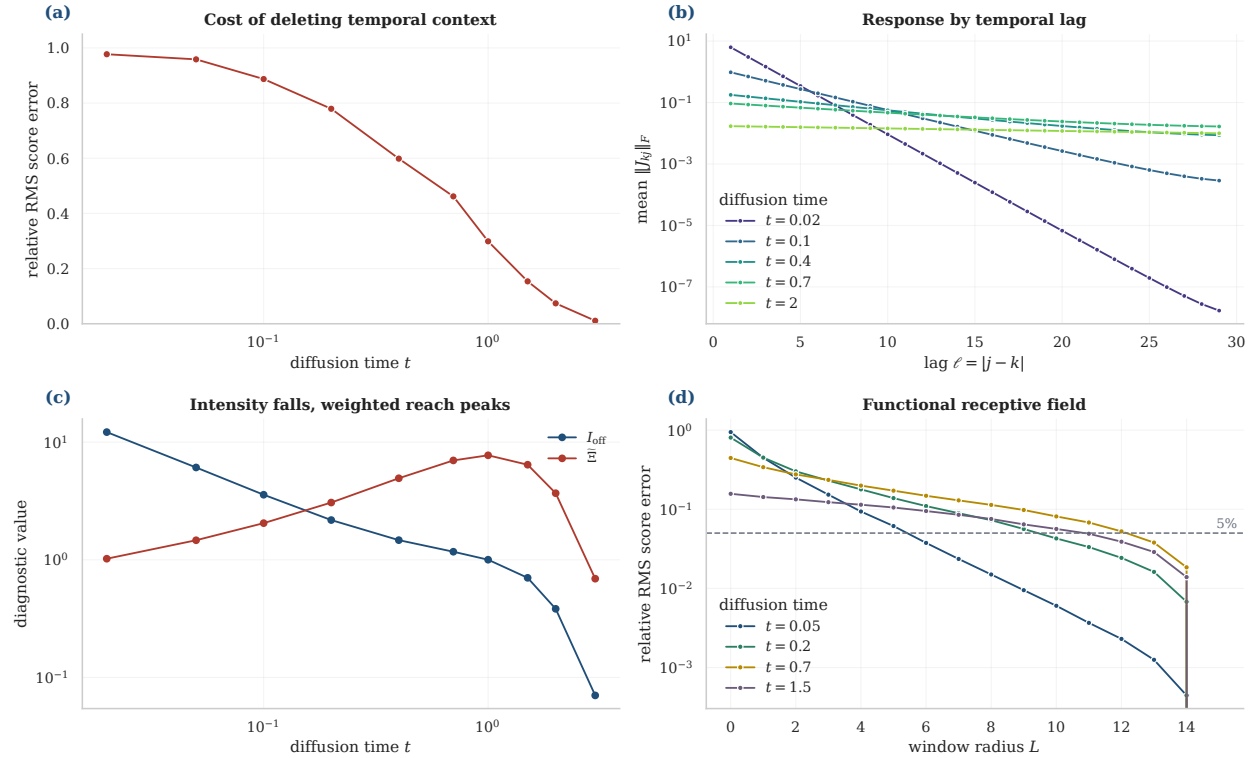


Figure 8. The exactly solvable surrogate. (a) Relative RMS error from deleting all temporal context. (b) Response profiles by lag. (c) Intensity falls monotonically while the weighted reach $\tilde{\Xi}$ rises, peaks near $t = 1$, and falls again. (d) Window error against window radius; the dashed line is 5%.

Three readings.

The cost of dropping context is largest at low noise. At $t = 0.02$ the per-frame marginal score has a relative RMS error of 0.977 — it is almost entirely wrong. The error falls to 0.154 at $t = 1.5$ and 0.011 at

$t = 3$. The decline does not mean the joint structure was unimportant at small t ; it means that at large t the score itself is approaching the universal Gaussian field $-x$, which the marginal model also reproduces.

Reach and intensity move oppositely. The intensity I_{off} falls by a factor of 173 between $t = 0.02$ and $t = 3$, while the weighted mean lag rises from 1.94 to 13.9 frames — that is, from a genuinely local profile to one close to flat, the structureless ceiling being 15. The composite $\tilde{\Xi}$ resolves the two: it rises to 7.7 at $t = 1$ and then falls, which is the behaviour one wants from a single summary.

The receptive field is non-monotone. For a 5% criterion the required window radius is $L = 6, 10, 13, 11$ at $t = 0.05, 0.20, 0.70, 1.50$. Context becomes more spread out at intermediate noise, and then stops mattering at all: the peak of $L_{5\%}$ at $t = 0.7$ sits close to the peak of $\tilde{\Xi}$, and neither is captured by ξ^{norm} , which increases monotonically over the whole range.

Scope of Part I

This is one parameter set of a model whose later frames are not radially confined. The qualitative mechanism it exposes — local clean score, nonlocal noised score, weak but broad at high noise — should be checked against the process that was actually drawn on the board. That is Part II.

Part II

The board problem

7 The polar process

7.1 Polar coordinates, and what we are trying to do

The board model is naturally written in polar coordinates, so we fix them first. A point $A \in \mathbb{R}^2$ is described by its distance from the origin and the angle it makes with the first axis,

$$A = r (\cos \theta, \sin \theta), \quad r = \|A\| \geq 0, \quad \theta \in [0, 2\pi), \quad (70)$$

and at each point the plane carries a natural moving orthonormal frame,

$$e_r = (\cos \theta, \sin \theta) \quad (\text{radial, pointing away from the origin}), \quad e_\theta = (-\sin \theta, \cos \theta) \quad (\text{tangential}). \quad (71)$$

e_θ is e_r rotated by a quarter turn, $e_\theta = R_{\pi/2} e_r$, so $\{e_r, e_\theta\}$ is orthonormal at every θ . Two facts are used repeatedly: moving along e_r changes the radius at fixed angle, moving along e_θ changes the angle at fixed radius; and the area element is $dx dy = r dr d\theta$, which is why a density in (r, θ) and a density in (x, y) differ by a factor r .

The decomposition matters because the model treats the two directions very differently: the radius is pulled back towards 1, while the angle drifts steadily and diffuses freely. The two channels are therefore expected to carry information over different distances, and one aim of the experiment is to measure that difference.

Figure 9 states the problem in these terms. Panels (a)–(c) are the model: polar coordinates, the confining potential, one clean trajectory. Panel (d) is the goal: given the whole *noisy* trajectory, the score block at frame k is a posterior estimate of the clean frame A_k , and we want to know which other frames contribute to it, how strongly, and from how far.

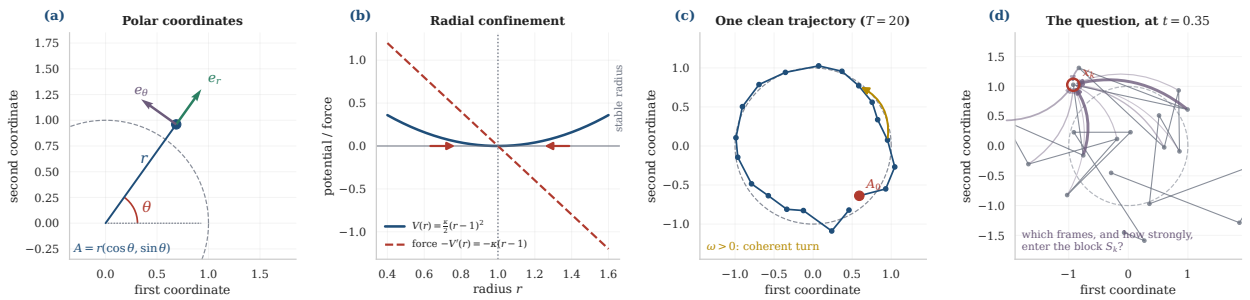


Figure 9. The problem. (a) Polar coordinates and the moving frame (71). (b) The quadratic potential $V(r) = \frac{\kappa}{2}(r-1)^2$ and the force $-V'(r)$ it generates; the force changes sign at $r = 1$, which is therefore the stable radius. (c) One clean trajectory of (72)–(74) at the baseline parameters: a noisy, coherently turning path near the unit circle. (d) The same trajectory after the channel (18) at $t = 0.35$. The score block S_k at the circled frame is a posterior estimate of the clean A_k built from *all* the noisy frames; arrow widths indicate the question, not the answer.

7.2 The equations

The clean dynamics run in internal time u :

$$dR_u = -\kappa(R_u - 1) du + \sqrt{2D_r} dB_u^{(r)}, \quad (72)$$

$$d\Theta_u = \omega du + \sqrt{2D_\theta} dB_u^{(\theta)}, \quad (73)$$

$$A_u = R_u (\cos \Theta_u, \sin \Theta_u), \quad (74)$$

with independent Brownian motions. We take the simplest constant-angular-drift reading of the board; [Section 13](#) records the alternatives.

7.3 Why the radial drift has this form

[Equation \(72\)](#) is the overdamped Langevin equation (7) for the quadratic potential

$$V(r) = \frac{\kappa}{2}(r - 1)^2. \quad (75)$$

A potential is a stored energy, and the force it generates is minus its slope,

$$F(r) = -V'(r) = -\kappa(r - 1). \quad (76)$$

The minus sign is the whole content: a system pushed by $-V'$ moves *downhill* in V , because then $\frac{d}{du}V(r_u) = V'(r)\dot{r} = -V'(r)^2 \leq 0$. Reading off the two cases in (76): if $r > 1$ then $F < 0$ and the force points inward; if $r < 1$ then $F > 0$ and it points outward. The radius is driven towards 1 from either side, at a rate proportional to the displacement, with κ setting the stiffness. [Figure 9\(b\)](#) draws both curves. Balancing this force against the noise gives the stationary radial law $\mathcal{N}(1, D_r/\kappa)$, by (9) with V from (75) and $k_B T \mapsto D_r$.

7.4 Exact radial transition

We need the law of R_{k+1} given R_k over one internal-time step h . Because (72) is linear, this can be obtained exactly, by the same integrating-factor argument as [Section 2.4](#).

Toolbox 7. The exact one-step radial law

Step 1: shift to the equilibrium radius. Put $Y_u = R_u - 1$. Since 1 is a constant, $dY_u = dR_u$, and (72) becomes a plain OU equation with no offset:

$$dY_u = -\kappa Y_u du + \sqrt{2D_r} dB_u. \quad (77)$$

Step 2: multiply by the integrating factor $e^{\kappa u}$. Consider $\phi(y, u) = e^{\kappa u} y$. This is linear in y , so $\partial_y^2 \phi = 0$ and by (5) Itô's lemma reduces to the ordinary product rule — there is no correction term:

$$d(e^{\kappa u} Y_u) = \underbrace{\kappa e^{\kappa u} Y_u du}_{\partial_u \phi du} + \underbrace{e^{\kappa u} dY_u}_{\partial_y \phi dY} \quad (78)$$

$$\begin{aligned} &= \kappa e^{\kappa u} Y_u du + e^{\kappa u} (-\kappa Y_u du + \sqrt{2D_r} dB_u) \\ &= \sqrt{2D_r} e^{\kappa u} dB_u. \end{aligned} \quad (79)$$

The two drift terms cancel by construction: that is what the factor $e^{\kappa u}$ is chosen for. What remains is an exact differential whose right-hand side contains no Y , so it can be integrated directly.

Step 3: integrate over one step. Integrate (79) from u_k to $u_{k+1} = u_k + h$. The left side telescopes:

$$e^{\kappa u_{k+1}} Y_{k+1} - e^{\kappa u_k} Y_k = \sqrt{2D_r} \int_{u_k}^{u_{k+1}} e^{\kappa s} dB_s. \quad (80)$$

Multiply throughout by $e^{-\kappa u_{k+1}}$. On the left, $e^{-\kappa u_{k+1}} e^{\kappa u_k} = e^{-\kappa(u_{k+1}-u_k)} = e^{-\kappa h}$; on the right, $e^{-\kappa u_{k+1}}$ is a constant with respect to the integration variable s and may be carried inside the integral:

$$Y_{k+1} = e^{-\kappa h} Y_k + \sqrt{2D_r} \int_{u_k}^{u_{k+1}} e^{-\kappa(u_{k+1}-s)} dB_s. \quad (81)$$

The first term is deterministic given Y_k ; the second is a stochastic integral of a *deterministic* kernel that weights the noise injected at time s by how much of it has survived by u_{k+1} .

Step 4: the stochastic integral has mean zero. This is (2) of Section 1.2. In the defining sum, each term is $f(s_i) (B_{s_{i+1}} - B_{s_i})$ with $f(s) = e^{-\kappa(u_{k+1}-s)}$ a fixed number and the increment independent of everything up to s_i with mean zero, so every term has mean zero and so does the limit. Consequently $\mathbb{E}[Y_{k+1} | Y_k] = e^{-\kappa h} Y_k$: the conditional mean simply relaxes geometrically towards 0, i.e. R relaxes towards 1.

Step 5: its variance, by Itô isometry. Apply (3) with the same deterministic f , then substitute $v = u_{k+1} - s$, so $dv = -ds$ and the limits $s = u_k, u_{k+1}$ become $v = h, 0$:

$$\text{Var} = 2D_r \int_{u_k}^{u_{k+1}} e^{-2\kappa(u_{k+1}-s)} ds = 2D_r \int_0^h e^{-2\kappa v} dv = 2D_r \cdot \frac{1 - e^{-2\kappa h}}{2\kappa} = \frac{D_r}{\kappa} (1 - e^{-2\kappa h}). \quad (82)$$

Since the kernel is deterministic, the integral is moreover exactly Gaussian (Section 1.2).

Step 6: name the constants and undo the shift. Set

$$a = e^{-\kappa h}, \quad q_r = \frac{D_r}{\kappa} (1 - a^2), \quad (83)$$

and return to $R = 1 + Y$:

$$R_{k+1} | R_k = r \sim \mathcal{N}(1 + a(r - 1), q_r). \quad (84)$$

□

Two checks. Because $a \in (0, 1)$, the conditional mean $1 + a(r - 1)$ lies strictly between r and 1: one step moves part of the way home. And iterating (84) gives stationary variance $q_r / (1 - a^2) = D_r / \kappa$, which is the Boltzmann value (9) for the potential (75) — the sampled chain and the continuous process agree.

7.5 Exact angular transition

Toolbox 8. The exact one-step angular law, and why it wraps

Step 1: integrate directly. The drift in (73) does not depend on Θ , so no integrating factor is needed. Integrating both sides over $[u_k, u_{k+1}]$ term by term,

$$\Theta_{k+1} - \Theta_k = \int_{u_k}^{u_{k+1}} \omega \, du + \sqrt{2D_\theta} \int_{u_k}^{u_{k+1}} dB_u^{(\theta)} = \omega h + \sqrt{2D_\theta} (B_{u_{k+1}}^{(\theta)} - B_{u_k}^{(\theta)}). \quad (85)$$

The first integral is elementary. The second is a stochastic integral with integrand 1, and by (1) it collapses to the single increment $B_{u_{k+1}} - B_{u_k}$, because every partial sum telescopes.

Step 2: the increment. By the definition of Brownian motion an increment over an interval of length h is $\mathcal{N}(0, h)$, independent of the past. Multiplying by the constant $\sqrt{2D_\theta}$ multiplies the variance by $2D_\theta$, so on the real line

$$\Theta_{k+1} | \Theta_k = \theta \sim \mathcal{N}(\theta + \omega h, q_\theta), \quad q_\theta = 2D_\theta h. \quad (86)$$

(The Itô isometry gives the same variance: $2D_\theta \int_{u_k}^{u_{k+1}} 1^2 \, ds = 2D_\theta h$.)

Step 3: wrap. What is observed is the point $A_u = R_u(\cos \Theta_u, \sin \Theta_u)$, which depends on Θ_u only modulo 2π . Reducing (86) modulo 2π and collecting all the windings that map to the same observed angle,

$$\Pr[\Theta_{k+1} \bmod 2\pi \in d\theta'] = \sum_{n \in \mathbb{Z}} \Pr[\Theta_{k+1} \in d\theta' + 2\pi n],$$

gives the *wrapped normal* law

$$\Theta_{k+1} | \Theta_k = \theta \sim \text{WN}(\theta + \omega h, q_\theta), \quad M_\theta(\theta' | \theta) = \sum_{n \in \mathbb{Z}} \frac{1}{\sqrt{2\pi q_\theta}} \exp\left[-\frac{(\theta' - \theta - \omega h + 2\pi n)^2}{2q_\theta}\right]. \quad (87)$$

When $q_\theta \ll (2\pi)^2$ a single winding dominates and the wrapped normal is indistinguishable from (86); the sum matters only when the angular uncertainty becomes comparable to a full turn. $\square$

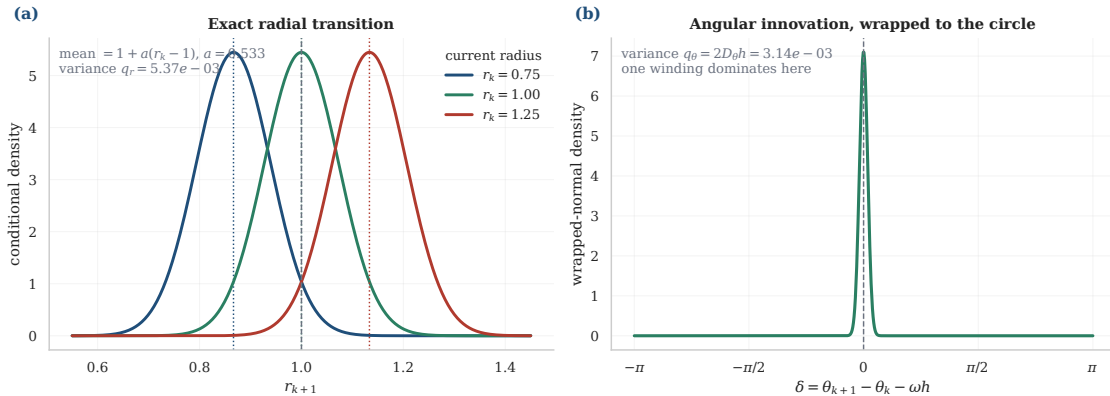


Figure 10. The two one-step transition laws at the baseline parameters. (a) Different current radii are pulled towards 1; dotted lines are the conditional means $1 + a(r_k - 1)$ of (84). (b) The angular innovation (87). Since $q_\theta = 3.1 \times 10^{-3}$ is far below $(2\pi)^2$, the wrapping is invisible here and the turn is sharply concentrated on ωh .

7.6 The clean law is Markov, and its score is local

With $H_k = (R_k, \Theta_k)$ the clean latent law factorises over one-step kernels,

$$p_0(h_{0:T-1}) = p_{\text{init}}(h_0) \prod_{k=0}^{T-2} M_r(r_{k+1} | r_k) M_\theta(\theta_{k+1} | \theta_k), \quad (88)$$

with M_r from (84) and M_θ from (87). This is the precise sense in which the clean trajectory is Markov, and it is the structure the research question asks us to exploit. We initialise at the stationary radial law $R_0 \sim \mathcal{N}(1, D_r/\kappa)$ and a uniform phase $\Theta_0 \sim \text{Unif}[0, 2\pi)$.

Differentiating (88) gives a local clean score, by the same two-incident-edges counting as in Section 5. With the radial innovation $\varepsilon_k^{(r)} = R_{k+1} - 1 - a(R_k - 1)$,

$$\frac{\partial}{\partial R_k} \log p_0 = -\frac{\varepsilon_{k-1}^{(r)}}{q_r} + a \frac{\varepsilon_k^{(r)}}{q_r} \quad (89)$$

for an interior frame: the first term checks consistency with where the previous radius left off, the second checks the prediction for the next. For the angle, with the wrapped innovation $\delta_k = \text{wrap}(\Theta_{k+1} - \Theta_k - \omega h)$ and $s_q(\delta) = \frac{d}{d\delta} \log \sum_n \exp[-(\delta + 2\pi n)^2/2q]$ (Appendix B),

$$\frac{\partial}{\partial \Theta_k} \log p_0 = s_{q_\theta}(\delta_{k-1}) - s_{q_\theta}(\delta_k). \quad (90)$$

When one winding dominates, $s_q(\delta) \approx -\delta/q$ and (90) becomes the familiar difference of neighbouring angular innovations.

Finally, to compare with the Cartesian score we must change variables. Because $dx dy = r dr d\theta$, the Cartesian density carries an extra factor $1/r$ relative to the polar one, and projecting the polar gradient onto the moving frame (71) gives

$$\nabla_{A_k} \log p_0 = e_{r,k} \left(\partial_{r_k} \log p_0 - \frac{1}{r_k} \right) + \frac{e_{\theta,k}}{r_k} \partial_{\theta_k} \log p_0. \quad (91)$$

The factor $1/r_k$ on the tangential component is the usual polar metric factor: a given change of angle moves the point farther when the radius is larger.

This clean score is local in internal time. The next section explains why the noised one is not.

8 The noised joint score

8.1 Why there is no closed form

At diffusion time t the channel (18) acts on each frame,

$$X_k | R_k, \Theta_k \sim \mathcal{N}\left(m_t R_k (\cos \Theta_k, \sin \Theta_k)^\top, \Delta_t I_2\right) =: g_t(x_k | h_k), \quad (92)$$

and the noised density is the integral over all latent trajectories,

$$p_t(x) = \int p_{\text{init}}(h_0) \prod_{k=0}^{T-2} M(h_{k+1} | h_k) \prod_{k=0}^{T-1} g_t(x_k | h_k) dh_{0:T-1}. \quad (93)$$

The integral has no simple global closed form, because the observation map

$$(r, \theta) \mapsto r(\cos \theta, \sin \theta)$$

is nonlinear and periodic. This is the one structural difference from Part I. There, the conditional law given the anchor was Gaussian with an a -independent covariance, so the latent integral collapsed to two dimensions. Here the latent state enters the likelihood through a nonlinear function, and no such collapse is available.

8.2 What Markovity buys instead

The factorisation (88) nonetheless makes (93) tractable, by replacing one $2T$ -dimensional integral with $2T$ two-dimensional ones. Define a forward function

$$\begin{aligned} f_0(h_0) &= p_{\text{init}}(h_0) g_t(x_0 | h_0), \\ f_{k+1}(h_{k+1}) &= g_t(x_{k+1} | h_{k+1}) \int M(h_{k+1} | h_k) f_k(h_k) dh_k, \end{aligned} \quad (94)$$

and a backward function

$$\begin{aligned} b_{T-1}(h_{T-1}) &= 1, \\ b_k(h_k) &= \int M(h_{k+1} | h_k) g_t(x_{k+1} | h_{k+1}) b_{k+1}(h_{k+1}) dh_{k+1}. \end{aligned} \quad (95)$$

The smoothed one-frame posterior is then $p(h_k | x) \propto f_k(h_k) b_k(h_k)$, and

$$\mathbb{E}[A_k | x] = \int r(\cos \theta, \sin \theta)^T p(r, \theta | x) dr d\theta, \quad (96)$$

after which the score follows from (22) and the response from (30).

How clean Markovity is exploited

The noised score can depend on every frame, but distant information cannot jump directly to frame k : it is transmitted through a sequence of one-step transition integrals. Two things follow. Computationally, the cost is linear in T rather than exponential. Structurally, the decay of the response with lag is a property of the one-step kernel, which is why the parameter sweeps of [Section 10](#) can predict it.

Numerical representation. The integrals are evaluated on a polar grid, $r \in [0.45, 1.55]$ with 41 points and $\theta \in [0, 2\pi)$ with 128 points for the response study. The radial interval spans more than six stationary standard deviations. Each one-step kernel is normalised on the grid, and the forward and backward functions are rescaled at every step to avoid underflow. The resulting score is exact for the discretised latent state space; [Section 11](#) measures the remaining grid error.

8.3 Looking at the score

The full score lives in $\mathbb{R}^{2T}$ and cannot be drawn. To see block S_k we fix every noisy frame except x_k and vary $x_k \in \mathbb{R}^2$; since $\nabla_{x_k} \log p_t(x_k | x_{-k}) = S_k(x, t)$, this is an exact two-dimensional conditional slice, not an approximation.

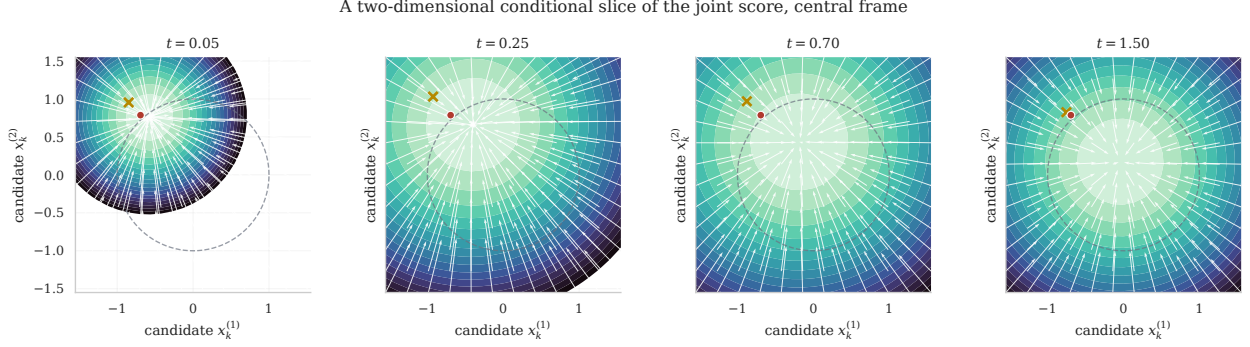


Figure 11. Conditional slices of the joint score at the central frame. Contours are $\log p_t(x_k | x_{-k})$, arrows the score. The red dot is the clean frame, the gold cross its noisy observation. At small t the other frames identify a specific phase on the ring; as t grows the field smooths out and approaches the Gaussian field $-x_k$.

The joint score localises a phase; the one-frame marginal keeps the ring symmetry

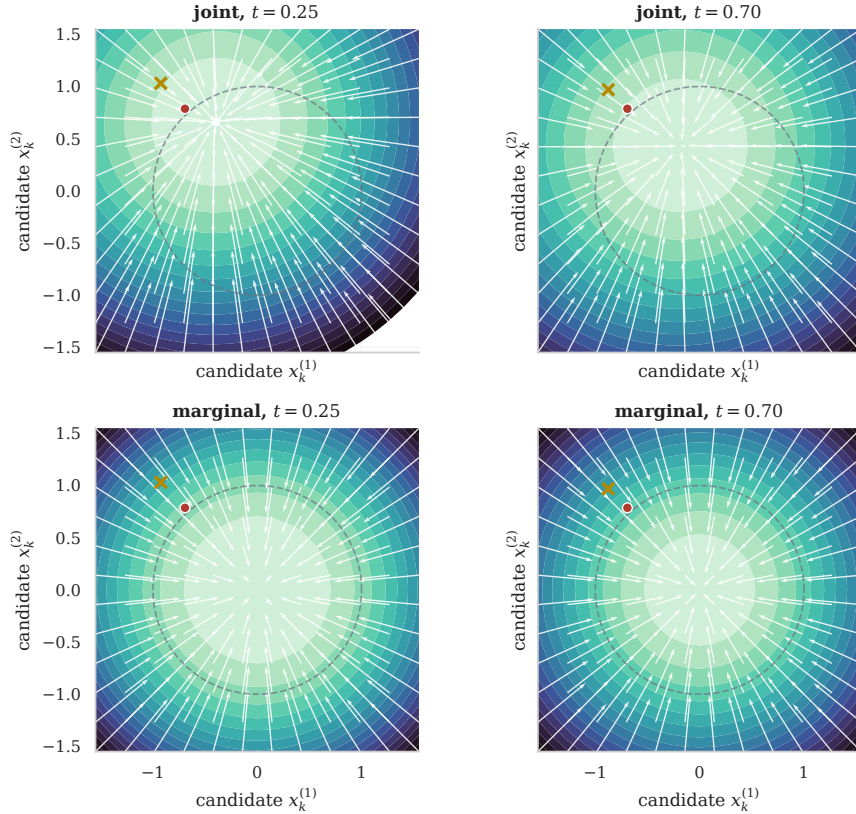


Figure 12. The two objects of (24), drawn. Top: the joint conditional field is localised near the phase inferred from the rest of the trajectory. Bottom: the one-frame marginal field stays close to ring-symmetric, because one frame alone does not identify where along the ring the trajectory is. This is the picture behind Figure 2.

9 A local linear model

The grid calculation is exact but opaque. A first-order expansion around a single rotating branch turns the model into a linear Gaussian state-space model, which can be read analytically. We derive it, and then state carefully when it may be used.

9.1 The expansion, order by order

Take the deterministic reference path $\bar{R}_k = 1$, $\bar{\Theta}_k = \theta_0 + k\omega h$ and write the deviations as

$$R_k = 1 + \rho_k, \quad \Theta_k = \bar{\Theta}_k + \varphi_k, \quad (97)$$

with $e_{r,k}, e_{\theta,k}$ the moving frame (71) evaluated at $\bar{\Theta}_k$. Rotating the reference direction by φ_k ,

$$(\cos \Theta_k, \sin \Theta_k) = \cos \varphi_k e_{r,k} + \sin \varphi_k e_{\theta,k}, \quad (98)$$

so that, exactly,

$$A_k = (1 + \rho_k)(\cos \varphi_k e_{r,k} + \sin \varphi_k e_{\theta,k}). \quad (99)$$

Now insert $\cos \varphi = 1 - \frac{\varphi^2}{2} + O(\varphi^4)$ and $\sin \varphi = \varphi - \frac{\varphi^3}{6} + O(\varphi^5)$ and expand:

$$A_k = \underbrace{e_{r,k} + \rho_k e_{r,k} + \varphi_k e_{\theta,k}}_{\text{first order}} - \underbrace{\frac{1}{2}\varphi_k^2 e_{r,k} + \rho_k \varphi_k e_{\theta,k}}_{\text{second order}} + O(\|(\rho_k, \varphi_k)\|^3). \quad (100)$$

Keeping the first order only:

Local Cartesian linearisation

$$A_k \approx e_{r,k} + \rho_k e_{r,k} + \varphi_k e_{\theta,k} \quad (101)$$

Radial deviations move perpendicular to the ring, angular deviations along the tangent, and to this order the two are decoupled. The two discarded terms have clear meanings: $-\frac{1}{2}\varphi^2 e_r$ is the curvature of the circle, which the tangent plane cannot see, and $\rho\varphi e_\theta$ is the fact that at larger radius a given angular deviation moves the point farther.

The truncation error is therefore quadratic. For a pure angular deviation at $\rho = 0$,

$$\|A_k - \hat{A}_k\| = \|(\cos \varphi - 1, \sin \varphi - \varphi)\| = \frac{\varphi^2}{2} + O(\varphi^3), \quad (102)$$

and keeping the second order reduces it to $\varphi^3/6$. Figure 13(a,b) shows both.

9.2 The linear Gaussian model it produces

Two of the three ingredients are already exactly linear. The radial deviation follows (84) verbatim,

$$\rho_{k+1} = a\rho_k + \sqrt{q_r} \xi_k^{(r)}, \quad (103)$$

an AR(1) chain, and on one unwrapped angular branch (86) gives a random walk,

$$\varphi_{k+1} = \varphi_k + \sqrt{q_\theta} \xi_k^{(\theta)}. \quad (104)$$

Together with the linear observation (101) and the linear channel (18), the whole trajectory is jointly Gaussian, $X_t \approx \mathcal{N}(\mu_t, \Sigma_t)$, and its score is explicit:

Taylor score

$$S^{\text{lin}}(x, t) = -\Sigma_t^{-1}(x - \mu_t). \quad (105)$$

9.3 When the expansion can hold, and when it cannot

Equation (102) makes the condition quantitative: the linearisation is accurate while the angular deviation from the reference branch stays small, and by (102) a 5% error in units of the ring radius corresponds to $\varphi \approx 0.32$ rad. Two different quantities must be checked against that number.

The clean spread is small. The angular deviation of the clean process is the random walk (104), so after $T - 1$ steps its standard deviation is $\sqrt{q_\theta(T - 1)}$. At the baseline, $q_\theta = 2D_\theta h = 3.14 \times 10^{-3}$ and $T = 20$ give 0.244 rad. The clean trajectory therefore stays within roughly a quarter of a radian of a single rotating branch over its whole length, and expanding around one branch is reasonable *for the prior*.

The posterior spread is what actually matters. The score involves the posterior of Θ_k given the noisy trajectory, and that spreads as t grows. Measuring it directly — the circular standard deviation of the smoothed angular posterior at the central frame, averaged over six trajectories — gives 0.085 rad at $t = 0.03$, 0.25 rad at $t = 0.4$, 0.40 rad at $t = 0.7$ and 1.62 rad at $t = 2$. The threshold 0.32 rad is crossed near $t \approx 0.55$, and Figure 13(c) shows the measured score error beginning to grow just after it.

The limits of one tangent plane

The expansion is local and phase-conditioned. It cannot represent a posterior spread over a large arc, several wrapped angular branches, or a multimodal phase. Beyond the crossing point these are not small effects to be bounded but qualitatively absent features, and the honest procedure is to compare against the nonlinear grid calculation — which is what Table 5 does. A second caveat is independent of t : the reference phase θ_0 must be estimated from the data, and the resulting error does not vanish as $t \rightarrow 0$. It is visible in Table 5 as a floor of 3–4% on the central-block error even at the smallest diffusion times, where the angular spread is negligible.

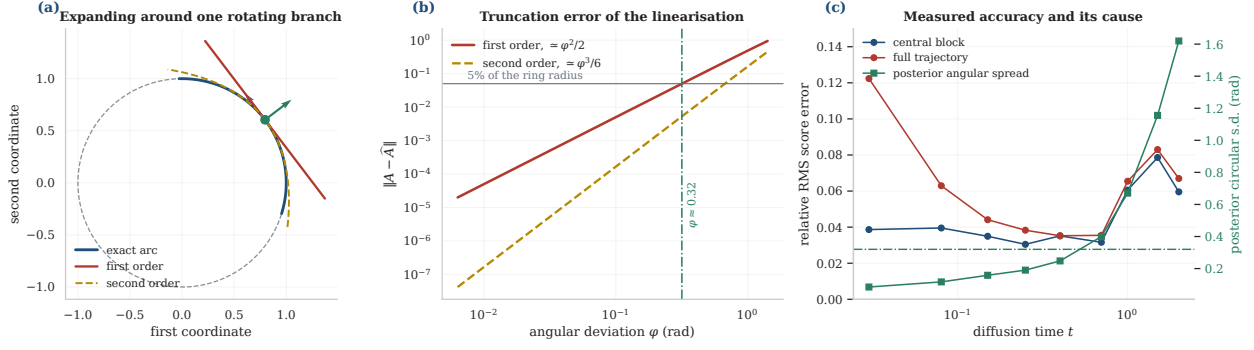


Figure 13. The linearisation and its validity. (a) Exact arc, first-order tangent, and second-order parabola through the reference point, with the moving frame. (b) Truncation error against angular deviation: first order is $O(\varphi^2)$ and second order $O(\varphi^3)$, so a 5% error corresponds to $\varphi \approx 0.32$ rad. (c) Measured relative score error of (105) against the nonlinear grid score (left axis), together with the measured circular standard deviation of the smoothed angular posterior (right axis). The error starts to grow shortly after the posterior spread crosses the threshold.

9.4 Analytic reach in the local model

The value of (105) is that its response can be computed by hand, which turns the vague statement “the one-step dynamics control the reach” into a formula. For an infinite interior chain, the AR(1) prior (103) has Fourier-space precision

$$Q_r(\varpi) = \frac{1 + a^2 - 2a \cos \varpi}{q_r}, \quad (106)$$

and the local observation (92) contributes precision $\lambda_t = m_t^2/\Delta_t$ at every frame. The posterior covariance in Fourier space is therefore $\widehat{C}_r(\varpi; t) = (\lambda_t + Q_r(\varpi))^{-1}$, and defining γ_r by

$$\cosh \gamma_r = \frac{1 + a^2 + q_r \lambda_t}{2a}, \quad (107)$$

the inverse transform is a pure exponential,

$$C_r(\ell, t) = \frac{q_r}{2a \sinh \gamma_r} e^{-\gamma_r |\ell|}, \quad \xi_r^{\text{lin}}(t) = \gamma_r^{-1}. \quad (108)$$

The angular channel is the $a \rightarrow 1$ case,

$$\cosh \gamma_\theta = 1 + \frac{q_\theta \lambda_t}{2}, \quad C_\theta(\ell, t) = \frac{q_\theta}{2 \sinh \gamma_\theta} e^{-\gamma_\theta |\ell|}, \quad (109)$$

and by the response identity (30) the score responses are

$$J_r(\ell, t) = \frac{m_t^2}{\Delta_t^2} C_r(\ell, t), \quad J_\theta(\ell, t) = \frac{m_t^2}{\Delta_t^2} C_\theta(\ell, t). \quad (110)$$

Equation (110) separates reach from amplitude explicitly, and in opposite directions. As t grows, $\lambda_t \rightarrow 0$, so $\gamma \rightarrow 0$ and the covariance length *increases*; at the same time the prefactor $m_t^2/\Delta_t^2 \rightarrow 0$, so the score response *vanishes*. This is the mechanism that Section 3.3 was built to summarise, now visible in closed form.

Clean correlation scales. For orientation, the clean process has stationary radial covariance $\text{Cov}(\rho_k, \rho_{k+\ell}) = (D_r/\kappa)e^{-\kappa h|\ell|}$, hence $\xi_r^{\text{clean}} = 1/(\kappa h)$, and first-harmonic angular coherence $\mathbb{E}[e^{i(\Theta_{k+\ell} - \Theta_k)}] = e^{i\omega h\ell} e^{-D_\theta h|\ell|}$, hence $\xi_\theta^{\text{clean}} = 1/(D_\theta h)$. At the baseline $h = 2\pi/20 \approx 0.314$, so

$$\xi_r^{\text{clean}} \approx 1.6 \text{ frames}, \quad \xi_\theta^{\text{clean}} \approx 637 \text{ frames}.$$

The angular coherence length exceeds the length of the trajectory by more than a factor of 30: within one data point the phase is essentially a single coherent quantity, while the radius forgets itself after a couple of frames. This asymmetry is the prediction that [Section 10](#) tests.

10 Measurements

Protocol. $T = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, $\omega = 1$, $h = 2\pi/20$, so the stationary radial standard deviation is $\sqrt{D_r/\kappa} = 0.087$. Per diffusion time, 28 independent clean trajectories for the response study and 36 for the window study; central frame $k = 10$, so $L_{\max} = 10$ and the structureless ceilings are $\bar{\ell} = 5.5$ and $\xi^{\text{norm}} = 10$. The radial and tangential projections

$$J_{kj}^{rr} = e_{r,k}^\top J_{kj} e_{r,j}, \quad J_{kj}^{\theta\theta} = e_{\theta,k}^\top J_{kj} e_{\theta,j}, \quad \text{etc.}, \quad (111)$$

use the *simulated* clean angle as an oracle basis. That is a diagnostic convenience, not an ingredient of the score; [Section 13](#) returns to it.

Table 2. Original polar model: reach and weighted reach by channel, the cost of dropping context, and the window radii. $\bar{\ell}$ is the weighted mean lag (34), $\tilde{\Xi}$ the relative weighted reach (35), $L^{5\%}$ the receptive radius (36). The structureless ceiling for $\bar{\ell}$ is 5.5.

t	marg. RMSE	$\bar{\ell}_r$	$\bar{\ell}_\theta$	$\tilde{\Xi}_r$	$\tilde{\Xi}_\theta$	$L_r^{5\%}$	$L_\theta^{5\%}$
0.03	0.732	1.98	4.19	0.208	2.166	2	9
0.10	0.733	2.71	5.01	0.119	2.575	3	9
0.40	0.513	5.04	5.40	0.314	2.672	7	9
0.70	0.349	5.47	5.45	1.292	2.701	8	9
1.00	0.229	5.46	5.46	1.022	1.793	8	9
1.50	0.092	5.48	5.47	0.928	1.042	5	7
2.00	0.037	5.47	5.47	0.428	0.428	0	0

10.1 Joint versus marginal

Replacing the joint score by per-frame marginal scores costs a relative RMS error of 0.73 at $t = 0.03$. The error stays above 0.69 up to $t = 0.2$, then falls, passing below 0.10 only around $t = 1.5$. The nonlinear process therefore agrees with the surrogate that the effect is not an artefact of the Gaussian construction. [Figure 12](#) shows the mechanism: one frame alone does not identify where along the ring the trajectory is.

10.2 The radial and tangential channels differ

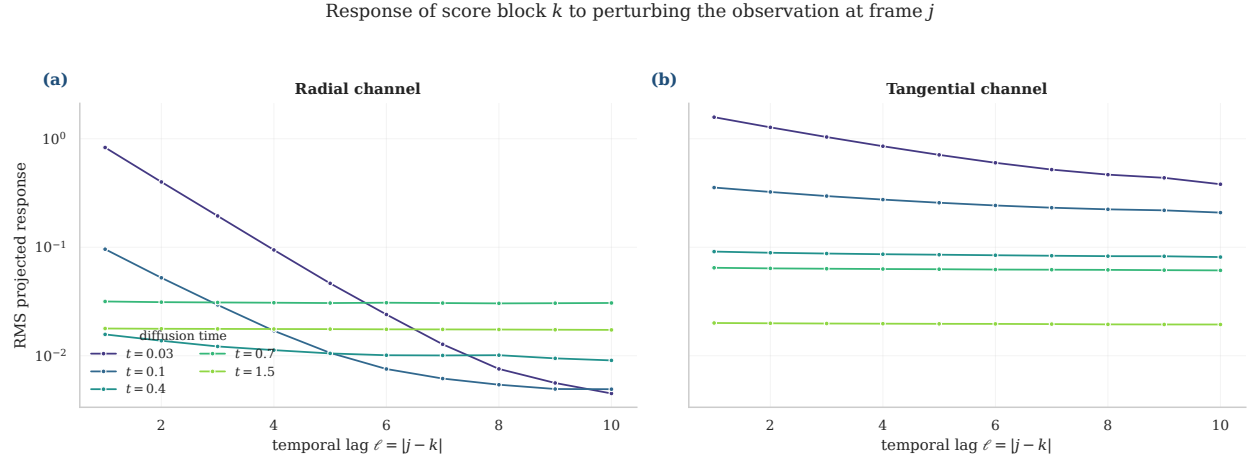


Figure 14. Projected response profiles (111). At small t the radial channel decays quickly with lag while the tangential channel is nearly flat across the whole trajectory. At larger t both profiles flatten, but their absolute levels are an order of magnitude lower.

At $t = 0.03$ the weighted mean lag is 1.98 frames radially and 4.19 frames tangentially, and the gap is much wider once intensity is included: $\tilde{\Xi}_r = 0.21$ against $\tilde{\Xi}_\theta = 2.17$, a factor of ten. The normalised range would have reported the same contrast as a factor of 2.5 only. This is the clean-scale prediction of Section 9.4 confirmed: radial fluctuations are damped by mean reversion after $\xi_r^{\text{clean}} \approx 1.6$ frames, while the phase is coherent over far more frames than the trajectory contains.

The local linear model is quantitatively right where it should be. At $t = 0.03$ it predicts a radial e-folding length of 1.35 frames, against a measured exponential fit of 1.39 frames — agreement to 3% (Table 3). Its tangential prediction, 4.45 frames, is already comparable to the chain, and the measured fit of 6.36 frames should be read as finite-size dominated rather than as an estimate of an infinite-system length.

Table 3. Low-noise exponential fits of the response profiles. These are reported for comparison with the analytic lengths of Section 9.4, and only at the three smallest diffusion times: beyond those the profiles are too flat for the fit to mean anything, and ξ_θ^{exp} exceeds 290 frames in a chain of 20. The bounded diagnostics of Section 3 are used everywhere else.

t	ξ_r^{exp}	R_r^2	ξ_θ^{exp}	R_θ^2
0.03	1.39	1.000	6.36	0.979
0.06	1.57	0.999	11.24	0.967
0.10	2.99	0.901	17.37	0.963

10.3 Reach and intensity move in opposite directions

Reach and strength move in opposite directions; the weighted diagnostic is non-monotone

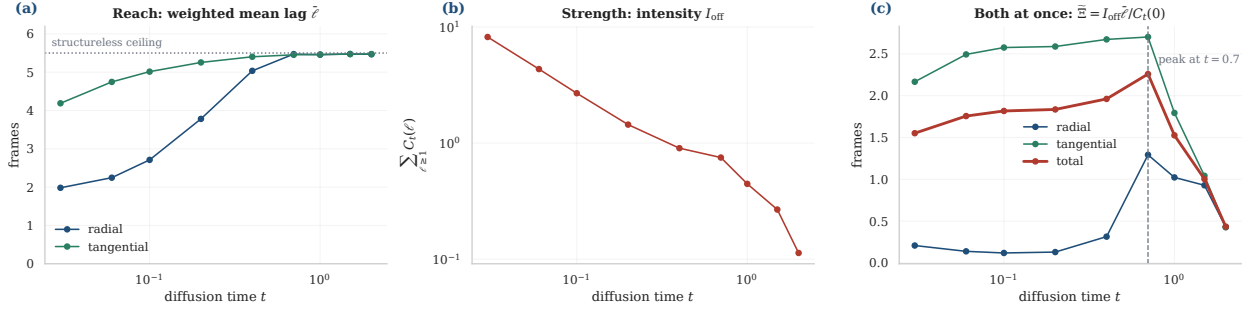


Figure 15. Reach, strength, and the two combined. (a) The weighted mean lag rises towards its structureless ceiling, the radial channel much later than the tangential one. (b) The absolute cross-frame intensity falls by a factor of about 73 over the same range. (c) The relative weighted reach $\tilde{\Xi}$ is non-monotone in all three channels, peaking at $t = 0.7$.

The two elementary diagnostics disagree, and both are right. Between $t = 0.03$ and $t = 2$ the intensity I_{off} falls from 8.2 to 0.11, a factor of about 73, while the weighted mean lag rises from 4.10 to 5.48 frames — that is, to within 0.4% of the structureless ceiling 5.5. The correct reading is not that high-noise frames are strongly coupled; it is that the weak residual coupling is spread evenly over the chain.

The composite $\tilde{\Xi}$ turns this into one number, and Figure 3(c) is the payoff. $\tilde{\Xi}$ rises from 1.55 at $t = 0.03$ to 2.26 at $t = 0.7$ and then falls to 0.43 at $t = 2$. The independently measured receptive radius $L_{5\%}$ does the same thing — 8, 8, 9, 9, 9, 6, 0 across $t = 0.03, \dots, 2$ — while ξ^{norm} rises monotonically over the whole range and reports nothing about the turn. Two remarks, in the interest of not over-claiming: the peaks do not coincide exactly (the surrogate has $\tilde{\Xi}$ peaking at $t = 1$ and $L_{5\%}$ at $t = 0.7$), and $\tilde{\Xi}$ is a first-moment summary, not a prediction of L_ϵ . What can be said is that both are non-monotone, that they turn in the same region, and that the two elementary diagnostics are monotone and therefore cannot indicate a turn at all.

10.4 The functional receptive field

Functional receptive field of the joint score

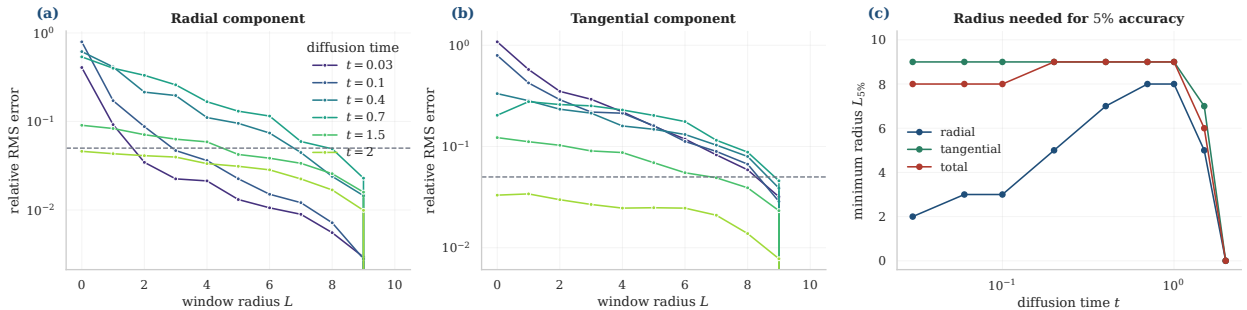


Figure 16. Window approximation of the joint score. (a,b) Relative RMS error of the windowed score against window radius, by channel; the dashed line is 5%. (c) The minimum radius $L_{5\%}$ needed for each channel. The radial score needs a radius of 2 at $t = 0.03$ but 8 at $t = 0.7$; the tangential score needs almost the whole trajectory at low noise, and nothing at all by $t = 2$.

At $t = 0.03$ the radial component of the score reaches 5% accuracy with a window radius of 2, while the tangential component needs 9 — nearly the whole trajectory. At intermediate diffusion times both channels need large windows. By $t = 2$ a single frame is already within 5% of the full score, because the cross-frame correction has become negligible; the required radius is 0. This is the same non-monotonicity as in the surrogate, and it is the diagnostic with the most direct implication for architecture: the useful temporal receptive field of a denoiser is a function of the noise level, and it is largest in the middle of the schedule.

10.5 The clean dynamics control the low-noise reach

The clean one-step dynamics set the low-noise response channels

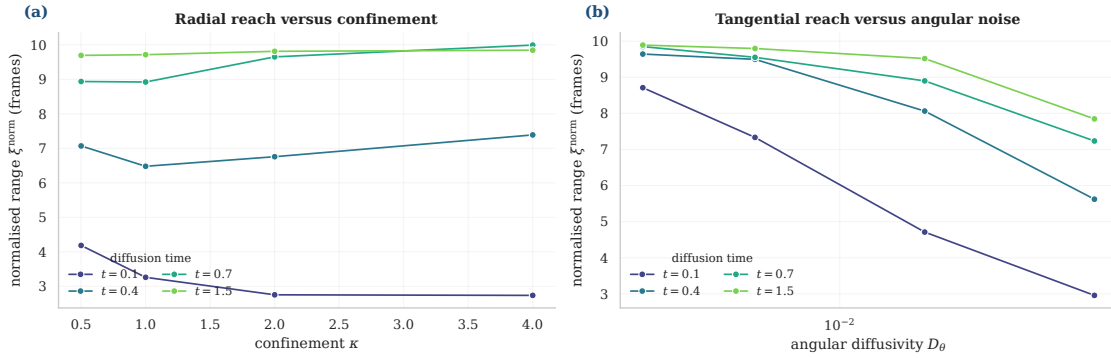


Figure 17. Sweeping the clean parameters. Stronger radial confinement κ shortens the radial reach; larger angular diffusivity D_θ shortens the tangential reach. At large diffusion time both curves are pushed towards the finite-size ceiling regardless of the parameter.

The low-noise trends follow the clean scales of Section 9.4. Increasing κ from 0.5 to 4 at $t = 0.1$ shortens the radial normalised range ξ_r^{norm} from 4.18 to 2.74 frames and reduces the nearest-neighbour radial intensity by an order of magnitude, from 0.35 to 0.034: stiffer confinement forgets faster. Increasing D_θ from 0.002 to 0.08 at the same diffusion time shortens ξ_θ^{norm} from 8.71 to 2.96 frames: faster dephasing destroys the long tangential channel. Both are what (107)–(109) predict, and together they are the concrete answer to the research question — the reach of the noised score is set by the one-step clean kernel, modulated by the observation precision λ_t .

11 Numerical validation

Every identity used above is checked against an independently computed reference.

Table 4. Validation summary. The full report is in `validation.json`.

check	discrepancy	tolerance
posterior covariance vs finite differences (polar)	6.63×10^{-9}	1.00×10^{-7}
analytic surrogate Jacobian vs finite differences	9.18×10^{-6}	1.00×10^{-4}
41×128 vs 51×192 response grid	3.04×10^{-11}	1.00×10^{-7}
rotation gauge round trip	2.22×10^{-16}	1.00×10^{-12}
clean score on an exact co-rotating ring path	0 (exact)	1.00×10^{-12}

Numerical check of the posterior response identity

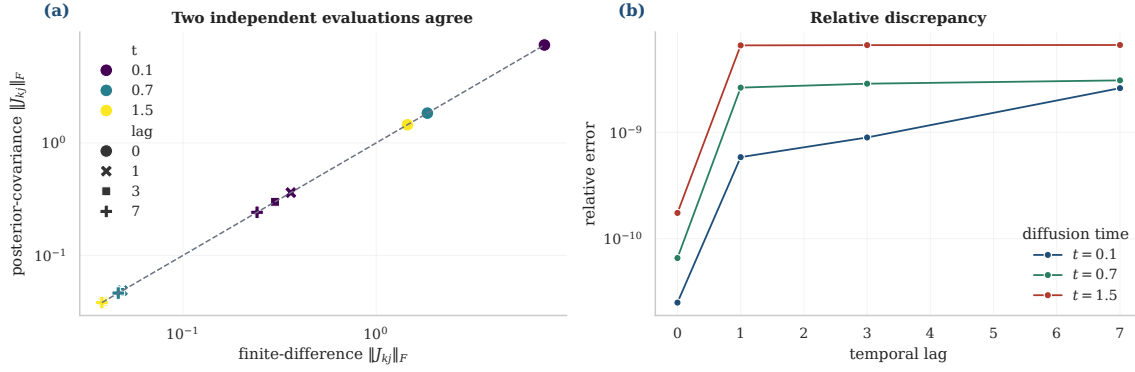


Figure 18. The response identity (30) evaluated two ways: from posterior pair covariances, and by central finite differences of the score after perturbing x_j . (a) The two agree along the diagonal. (b) Relative discrepancies stay below 6.7×10^{-9} in the cases tested.

Table 5. First-order Taylor score against the nonlinear grid score. The central-block error has a floor of 3–4% from the estimated reference phase; the growth beyond $t \approx 0.7$ follows the angular posterior spread (Section 9.3).

t	central RMSE	trajectory RMSE	mean cosine
0.03	0.039	0.122	0.985
0.15	0.035	0.044	0.997
0.40	0.035	0.035	0.998
0.70	0.032	0.035	0.999
1.00	0.061	0.065	0.992
1.50	0.079	0.083	0.990
2.00	0.060	0.067	0.991

Beyond the table, the checks include row normalisation of the transition kernels, global rotation equivariance of the score, stability under halving the finite-difference step, and the exact vanishing of the clean surrogate score on a perfect co-rotating ring path (Section 5, point 2).

One caveat on grid convergence. The difference between the 41×128 and 51×192 response grids is 3×10^{-11} , which is small because the baseline likelihoods and kernels are already very smooth at those resolutions. A deliberately coarse 11×24 grid does show visible deviations, so the production grid is not trivially grid-independent; it is converged for these parameters.

12 What the experiment answers

12.1 How Markovity helps, and how it does not

The clean law is local, $p_0(h_{0:T-1}) = p_0(h_0) \prod_k M(h_{k+1} | h_k)$, and this gives three concrete advantages: the clean score is assembled from adjacent transition innovations (Equations (89) and (90)); the noised posterior can be integrated by repeated one-step operators instead of one integral over $2T$ latent variables (Equations (94) and (95)); and any distant influence on S_k must propagate through a sequence of local transitions, which is why the parameter sweeps of Section 10 can predict the reach from the one-step kernel.

What clean Markovity does *not* give is locality of the noised score. For $t > 0$,

$$S_k(x, t) \neq F_t(x_{k-1}, x_k, x_{k+1})$$

in either model: integrating out uncertain latent states creates posterior correlations at every lag. In the surrogate this is explicit — the rank-one anchor term of (69) touches every pair of frames — and in the polar model it is measured.

12.2 The mechanism, in one line

Both models show the same sequence,

$$\text{strictly local clean score} \longrightarrow \text{broader noised posterior dependence} \longrightarrow \text{weak universal field } -x. \quad (112)$$

At intermediate diffusion times individual frames are noisy enough that distant context improves the estimate of a clean frame. At large diffusion times the same context is still statistically present but no longer materially useful, because the prefactor m_t^2/Δ_t^2 in (30) has gone to zero.

12.3 The radial–tangential separation

The polar process shows something the Cartesian surrogate cannot express: radial information is damped by mean reversion, while tangential information rides a slowly diffusing global phase. The two score channels therefore have different reaches and different receptive fields, predicted at low noise by $\xi_r^{\text{clean}} = 1/(\kappa h)$ and $\xi_\theta^{\text{clean}} = 1/(D_\theta h)$ and confirmed by the sweeps.

Answer to the research question

The clean Markov dynamics are exploited not by assuming that the noised score is local, but by identifying the local channels through which information propagates. The cross-frame response of the joint score is a posterior covariance (30); its radial and tangential decay scales are controlled by the one-step kernel through (107)–(109); and its practical receptive field depends on the diffusion time and is largest in the middle of the schedule. In the model and parameters studied here, tangential context is almost global near the data while radial context is initially local, and diffusion broadens both normalised profiles while weakening their absolute intensity until the score becomes effectively framewise again.

13 Limitations and next steps

1. **One parameter set.** Every number reported here comes from a single baseline, with $T = 20$ (polar) or $T = 30$ (surrogate) and 28–36 trajectories per diffusion time. Only κ and D_θ were swept. Nothing is claimed about scaling in T .
2. **Angular equation.** We used a constant angular drift. The board admits other readings; a different angular drift changes the transition kernel but not the score identities of Section 2.
3. **Radius positivity.** The shifted OU process (77) is formally supported on all of $\mathbb{R}$. At the baseline the probability of a negative radius is negligible, but a strict model would reflect at the origin or use a positive diffusion.

4. **Known rotation.** The gauge reduction of [Section 4.4](#) assumes ψ known. An unknown clockwise/counter-clockwise class would add a discrete latent variable and may produce a genuine speciation phenomenon of the kind discussed in [Section 1.6](#). This is the most interesting single extension.
5. **Finite trajectory.** At large t the reach diagnostics saturate at their structureless ceilings. This is a finite-size effect, and is not evidence of a divergent correlation length.
6. **Oracle projections.** The radial and tangential channels were separated using the simulated clean angle. A practical method would infer the phase, and the resulting error would enter the projected profiles.
7. **Grid approximation.** The polar score is numerically converged for these parameters, but it is not a symbolic closed form.
8. **One tangent plane.** The Taylor model cannot represent wrapped or multimodal angular posteriors, which by [Section 9.3](#) is the regime beyond $t \approx 0.55$.

The immediate extensions are to vary T systematically, to infer the angular phase rather than use an oracle basis, to add an unknown rotation class, and to test whether a time-dependent local architecture with receptive field $L_\epsilon(t)$ can match the full score.

A The anchor posterior in polar form

This appendix records, for reproducibility only, the one-dimensional integrals the code uses to tabulate (67). Nothing in the main argument depends on them.

Align the first axis with h and write $a = r(\cos \theta, \sin \theta)$, $da = r dr d\theta$, so that $h^\top a = \rho r \cos \theta$ with $\rho = \|h\|$. The four angular integrals needed are standard integral representations of the modified Bessel functions I_ν :

$$\int_0^{2\pi} e^{\rho r \cos \theta} d\theta = 2\pi I_0(\rho r), \quad \int_0^{2\pi} \cos \theta e^{\rho r \cos \theta} d\theta = 2\pi I_1(\rho r), \quad (113)$$

$$\int_0^{2\pi} \cos^2 \theta e^{\rho r \cos \theta} d\theta = \pi [I_0 + I_2](\rho r), \quad \int_0^{2\pi} \sin^2 \theta e^{\rho r \cos \theta} d\theta = \pi [I_0 - I_2](\rho r). \quad (114)$$

With $w(r) = \exp[-(r-1)^2/2\lambda - \beta r^2/2]$ they give

$$\mu(\rho) = \frac{\int_0^\infty r^2 w(r) I_1(\rho r) dr}{\int_0^\infty r w(r) I_0(\rho r) dr}, \quad (115)$$

and the parallel and perpendicular second moments follow from the third and fourth integrals, after which $\text{Cov}(A|x) = v_\perp I_2 + (v_\parallel - v_\perp) \widehat{h} \widehat{h}^\top$. All radial integrals are evaluated by Gauss–Legendre quadrature, with the exponential rescaled to avoid overflow.

B The exact wrapped angular score

With $G_q(\delta) = \sum_{n \in \mathbb{Z}} \exp[-(\delta + 2\pi n)^2/2q]$,

$$s_q(\delta) = \frac{G'_q(\delta)}{G_q(\delta)} = -\frac{\sum_n (\delta + 2\pi n) \exp[-(\delta + 2\pi n)^2/2q]}{q \sum_n \exp[-(\delta + 2\pi n)^2/2q]}. \quad (116)$$

This is periodic and smooth across the branch cut at $\pm\pi$. When q is small and δ is away from $\pm\pi$, the nearest winding dominates and $s_q(\delta) \approx -\delta/q$, which is the linear-Gaussian score used in (104).

C Reproduction

From the bundle root:

```
python code/compute_surrogate_full_windows.py # window sweep for the surrogate
python code/make_tables.py # all LaTeX tables from data/
python code/make_figures.py # all figures
python code/validate.py # numerical checks -> validation.json
tectonic -X compile main.tex # or latexmk -pdf main.tex
```

The curated CSV files under `data/` are the canonical numerical outputs; the heavier scripts that produced them are preserved as `code/run*_original.py`. No number in the text or tables is entered by hand: `code/make_tables.py` regenerates every table from `data/` and `validation.json`, and `code/metrics.py` contains the single definition of the diagnostics of Section 3.

Study	Main settings
Surrogate, exact	$T = 30$, $\lambda = 0.05$, $\sigma = 0.15$, 350 trajectories, quadrature anchor posterior
Polar, response	$T = 20$, $\kappa = 2$, $D_r = 0.015$, $D_\theta = 0.005$, 41×128 grid, 28 trajectories per diffusion time
Polar, windows	36 trajectories per diffusion time, radii $L = 0, \dots, 10$
Taylor comparison	32 trajectories per diffusion time; reference phase estimated by de-rotating and averaging the noisy frames
Angular posterior spread	6 trajectories per diffusion time, 41×128 grid
Score-field figures	one fixed clean trajectory and one fixed standardised noise realisation, 31×64 grid

Simulation provenance.

References

- J. Abramson et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* 630:493–500, 2024.
- B. Achilli, M. Benedetti, G. Biroli, and M. Mézard. Theory of speciation transitions in diffusion models with general class structure. arXiv:2602.04404, 2026.
- B. D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications* 12(3):313–326, 1982.
- D. Bachtis, G. Biroli, A. Decelle, and B. Seoane. Cascade of phase transitions in the training of energy-based models. *Advances in Neural Information Processing Systems* 37, 2024.
- G. Biroli, T. Bonnaire, V. De Bortoli, and M. Mézard. Dynamical regimes of diffusion models. arXiv:2402.18491, 2024.
- A. Einstein. Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen. *Annalen der Physik* 322(8):549–560, 1905.
- K. Itô. Stochastic integral. *Proceedings of the Imperial Academy of Tokyo* 20:519–524, 1944.
- L. D. Landau and E. M. Lifshitz. *Statistical Physics, Part 1*. Pergamon, 3rd edition, 1980.
- P. Langevin. Sur la théorie du mouvement brownien. *Comptes Rendus de l’Académie des Sciences* 146:530–533, 1908.

- M. Mézard. Generative diffusion: updated notes. Lecture notes, December 2025.
- M. Mézard and A. Montanari. *Information, Physics, and Computation*. Oxford University Press, 2009.
- R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- A. Sclocchi, A. Favero, and M. Wyart. A phase transition in diffusion models reveals the hierarchical nature of data. *Proceedings of the National Academy of Sciences*, 2025. arXiv:2402.16991.
- J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. *International Conference on Machine Learning*, 2015. arXiv:1503.03585.
- Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations*, 2021.
- G. E. Uhlenbeck and L. S. Ornstein. On the theory of the Brownian motion. *Physical Review* 36:823–841, 1930.